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PERSPECTIVE

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Transformative frontiers of photonics for next-generation optical communication networks

Alberto Boretti[1*]

*Correspondence:
Alberto Boretti
a.a.boretti@gmail.com
[1]Wellington, New Zealand

Abstract
Photonics, the science and technology of light, is undergoing a profound transformation, driving innovations that are critical for advancing optical communication networks. This perspective illuminates the key emerging frontiers poised to shape the future of global connectivity. We explore the convergence of integrated photonics, which miniaturizes optical systems onto chips for unprecedented performance and efficiency in transceivers and switches, and quantum photonics, which harnesses the fundamental quantum properties of light to revolutionize secure communications through quantum key distribution (QKD). We delve into the disruptive potential of metaphotonics, where engineered materials provide unparalleled control over light, enabling novel functionalities for spectral filtering, beam steering, and signal processing within network nodes. Furthermore, we examine the critical role of biophotonics in advancing distributed fiber sensor networks for the real-time monitoring of optical infrastructure health. Finally, we highlight the imperative of green photonics, which focuses on creating energy-efficient technologies essential for sustainable network operation. By synthesizing breakthroughs across these synergistic domains, this perspective offers a forward-looking vision of the challenges and immense opportunities for overcoming capacity and energy bottlenecks in future optical networks, charting a course for interdisciplinary collaboration and technological breakthroughs.

Keywords Integrated photonics, Quantum photonics, Metamaterials, Biophotonics, Green photonics, Optical metasurfaces, Optical networks

1 Introduction

From the dawn of civilization, light has been both a source of wonder and a fundamental tool shaping human progress. Initially harnessed for illumination and rudimentary signaling, our understanding and mastery of light have undergone a dramatic transformation, particularly in the last half-century, largely driven by the demands of global communications. This revolution has given rise to photonics, the science and engineering discipline dedicated to the generation, manipulation, and detection of light quanta, photons. The energy of a single photon is given by the Planck-Einstein relation:

$$E = h \times v = h \times c / \lambda$$

Where E is the energy of the photon, h is Planck's constant, ν is the frequency of the light, c is the speed of light, and λ is its wavelength. This fundamental equation underscores the quantum nature of light that is at the heart of photonics and dictates the photon energy at standard telecommunications wavelengths around 1550 nm (where $\lambda \approx 1.55 \times 10^{-6}$ m). Just as photons carry energy, they also possess momentum, a property that is fundamental to applications ranging from optical tweezers to solar sails. The momentum of a photon is inversely proportional to its wavelength:

$$p = h/\lambda$$

No longer confined to niche applications, photonics has ascended to become a truly enabling and pervasive technology, underpinning critical infrastructure and driving innovation across an astonishingly broad spectrum of sectors. From the invisible backbone of global communication networks—fiber optic cables transmitting vast quantities of data across continents [1]—to the intricate laser systems powering advanced manufacturing [2] and the sophisticated optical sensors revolutionizing environmental monitoring [3], photonics is the silent yet indispensable force propelling modern technological advancement [4]. The capacity of these optical links is fundamentally governed by the Shannon-Hartley theorem for a channel with bandwidth B and signal-to-noise ratio SNR,

$$C = B \log_2 (1 + SNR)$$

Which sets a theoretical limit that emerging photonic technologies strive to overcome.

The relentless pursuit of faster communication, more efficient computation, enhanced healthcare, and sustainable energy solutions has placed photonics at the forefront of scientific and technological endeavor. This perspective focuses on five pivotal domains within photonics that are currently experiencing exponential growth and exhibiting transformative potential, with a specific emphasis on their application to optical communication networks. These areas, poised to redefine the industry, are: integrated photonics, quantum photonics, metaphotonics and metamaterials, biophotonics and biomedical imaging, and green photonics for sustainability. Each of these areas represents not just a sub-discipline, but a vibrant frontier of innovation, driven by both fundamental scientific curiosity and the imperative to create impactful technologies for next-generation networks.

To further contextualize the developmental stage of each photonics domain from a networking perspective, we introduce a Technology Readiness Level (TRL) curve that maps each area's maturity against its potential impact on optical communications:

- Silicon Quantum Photonics [5] at TRL 4 with High Impact—reflecting its successful lab-scale demonstrations of programmable quantum circuits for QKD but need for further scaling to > 100 qubits for complex network integration.
- Meta-Lens for Beam Steering in Free-Space Optics [6] at TRL 5 with High Impact—owing to its potential for compact, non-mechanical beam steering in free-space optical communication links.
- Bio-based Photonic Glasses for Sustainable Fiber Manufacturing [7] at TRL 3 with Medium Impact—characterized by promising biodegradation results but unresolved challenges in thermal stability and cost reduction for telecom-grade fiber.

This curve highlights that while certain technologies like meta-lenses are advancing towards practical deployment (TRL 5–6), others such as quantum photonics (TRL 4) and green photonics (represented by bio-based glasses at TRL 3) require 3–5 more years of development to reach commercialization (TRL 8), **primarily due to scalability and compatibility barriers with existing network infrastructure.**

Integrated photonics emerges as a crucial paradigm shift in how we design and deploy optical systems for networking. Moving away from bulky, discrete optical components, integrated photonics seeks to replicate the success of microelectronics by miniaturizing optical circuits onto semiconductor chips. This integration promises to dramatically reduce the size, cost, and power consumption of photonic devices, while simultaneously enhancing their performance and scalability. This is essential for meeting the ever-increasing demands of data centers, telecommunications, and sensing applications where compactness and efficiency are paramount [8, 9]. The nonlinear Schrödinger equation, which governs signal propagation in optical fibers,

$$\partial A/\partial z \;+\; (\alpha/2)\,A \;+\; i\,(\beta_2/2)\,\left(\partial A/\partial T^2\right) - i\gamma|A|^2 A = \,0$$

Where A is the pulse amplitude, α is attenuation, β_2 is dispersion, and γ is the nonlinear parameter, must be carefully managed in integrated photonic circuits to minimize signal degradation.

Quantum photonics, on the other hand, delves into the fundamental quantum nature of light, harnessing phenomena such as superposition and entanglement to unlock entirely new capabilities for secure communications. A two-qubit system can exist in an entangled state, where the measurement outcome of one qubit is perfectly correlated with the other, regardless of the distance separating them. One of the four maximal entangled Bell states is represented as:

$$|\Phi^+\rangle = (1/\sqrt{2})\,(|00\rangle \;+\; |11\rangle)$$

This burgeoning field holds the key to revolutionary advancements in computing, promising to tackle problems intractable for even the most powerful classical computers. Furthermore, quantum photonics underpins the development of ultra-secure communication networks impervious to eavesdropping and highly sensitive sensors capable of detecting minute changes in physical parameters, pushing the boundaries of measurement and information processing [10, 11]. **A key challenge here lies in scaling quantum systems while maintaining photon purity—a tension between integration compatibility and quantum performance.** The security of QKD protocols like BB84 relies on the no-cloning theorem, which prevents an eavesdropper from perfectly copying a quantum state, thereby ensuring the integrity of the cryptographic key.

Metaphotonics and metamaterials represent a radical departure from traditional optics by engineering artificial materials with optical properties not found in nature. By structuring materials at the subwavelength scale, metaphotonics enables unprecedented control over light-matter interactions, allowing for the creation of novel optical components with functionalities that defy conventional limitations. From perfect lenses and invisibility cloaks to advanced optical filters and highly efficient nonlinear optical devices, metaphotonics is opening up entirely new horizons in light manipulation and application [12, 13], with direct relevance to creating ultra-compact multiplexers and routers for dense wavelength-division multiplexing (WDM) systems.

In the realm of biophotonics and biomedical imaging, light is being harnessed as a powerful tool for probing the intricate complexities of biological systems and revolutionizing healthcare. Optical techniques offer non-invasive and minimally invasive methods for diagnostics, enabling early disease detection, personalized medicine, and real-time monitoring of physiological processes. From advanced microscopy techniques providing unprecedented cellular-level detail to innovative phototherapeutic approaches for targeted cancer treatment and tissue regeneration, biophotonics is transforming our understanding of life and improving human health [14, 15]. These techniques can be adapted for sensing applications relevant to network infrastructure monitoring.

Finally, green photonics addresses the critical imperative of sustainability by focusing on developing energy-efficient and environmentally friendly photonic technologies. This encompasses a wide range of applications, from improving the efficiency of solar energy harvesting and solid-state lighting to developing photonic sensors for environmental monitoring and optimizing energy consumption in communication networks. The power consumption of a network, $P_{network}$, is a sum of its components,

$$P_{network} = \sum \left(P_{transreceiver} + P_{switch} + P_{amplifier}\right)$$

Green photonics is essential for mitigating the environmental impact of technology and paving the way for a more sustainable future [16, 17], and is critical for reducing the carbon footprint of the rapidly expanding global internet infrastructure. Figure 1 summarizes the photonics research domains covered in this perspective.

2 Materials and methods

Our approach involved a targeted literature search within each of these areas, prioritizing highly cited works to establish foundational concepts and landmark achievements, alongside recent publications from 2024 and 2025 that reflect the cutting edge of research. Specifically, for each subtopic, we analyzed approximately 60 works selected using Scopus database. This methodology ensures a balanced perspective, capturing both historical context and the latest breakthroughs.

Rather than offering an exhaustive survey, this perspective presents a representative snapshot of the most dynamic and impactful research directions. By synthesizing diverse

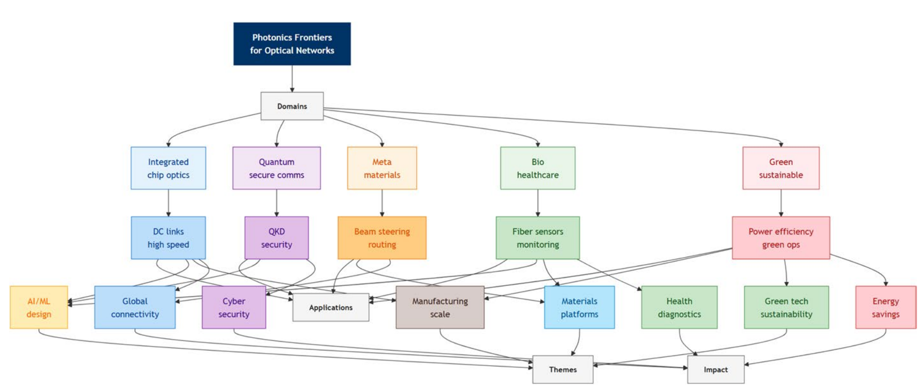

Fig. 1 Overall framework mapping photonics frontiers to optical networks, showing the linkage between core domains, key applications, enabling research themes, and societal impacts

advancements, we aim to uncover common themes, address key challenges, and provide insight into the future of photonics, encouraging fresh research avenues, cross-disciplinary partnerships, and technological developments that will shape the next era of innovation.

3 Integrated photonics

The relentless quest for smaller, faster, and more energy-efficient devices is a defining trend in modern technology. Integrated photonics, particularly silicon photonics, is emerging as a game-changer in this pursuit, promising to revolutionize a wide range of applications by bringing the power of light onto microchips [18, 19].

By fabricating photonic circuits directly onto silicon chips, similar to the fabrication processes used in microelectronics, integrated photonics offers a path to unprecedented scalability, cost-effectiveness, and miniaturization [20, 21]. This approach allows for the integration of numerous optical components, such as lasers, modulators, detectors, and waveguides, onto a single chip, creating complex and highly functional photonic systems. The explosive growth of data traffic, driven by cloud computing, streaming services, and the Internet of Things, places immense pressure on data centers. Integrated photonics, particularly silicon photonics, offers a compelling solution to the bandwidth bottlenecks in these facilities, providing high-speed, low-power interconnects to replace traditional copper wiring.

The exponential growth of cloud computing (15% annual data traffic increase) has created a bandwidth crisis in data center interconnects, where copper wiring limits speeds to < 100 Gbps. 3D integrated silicon transmitters [22] achieve 224 Gbaud by stacking photonic layers–- doubling data throughput vs. 2D designs. The data rate can be calculated as:

$$DataRate \ = \ BaudRate \ \times \ \log_2(M)$$

Where M is the number of symbols in the modulation format.

However, thermal crosstalk between layers causes > 5 °C temperature rises, requiring active cooling. This heat dissipation challenge is governed by fundamental heat transfer principles. The steady-state heat conduction across a layer is described by Fourier's Law:

$$q = -k \ \times \ A \ \times (dT/dx)$$

Where q is the heat flow rate, k is the thermal conductivity of the material, A is the cross-sectional area, and dT/dx is the temperature gradient.

Similarly, conventional grating couplers suffer from 2 dB insertion loss, wasting 37% of optical power in long-haul networks. The insertion loss (IL) in decibels (dB) is given by:

$$IL\,(dB) \ = \ 10 \ \times \ \log_{10}\,(Pin\,/\,Pout)$$

Where Pin is the input power and Pout is the output power. Inverse-designed vertical couplers [23] reduce loss to 0.8 dB via bottom reflectors. However, sensitivity to fabrication errors–- > 10% loss if feature size varies by > 50 nm.

The advantages of the high level of integration, compactness, robustness, and reduced cost extend far beyond the original telecommunications applications. A key metric for the performance of many resonant integrated photonic devices, such as micro-ring resonators used for filtering and sensing, is the Quality factor (Q-factor). The Q-factor represents the degree of energy confinement within the resonator, with higher values indicating lower losses and sharper resonance. It is defined as the ratio of the resonant frequency (or wavelength) to the resonance bandwidth:

$$Q = \omega_0 / \Delta\omega = \lambda_0 / \Delta\lambda$$

Integrated Photonics finds application in sensing, biophotonics, quantum computing, and automotive LiDAR. The versatility of integrated photonics circuits that can be designed, programmed and reconfigured also plays a key role [24] Integrated photonics, particularly silicon photonics, leverages the mature and highly developed manufacturing infrastructure of the CMOS (Complementary Metal–Oxide–Semiconductor) industry. This crucial advantage significantly reduces production costs, increases scalability, and enables mass production, making integrated photonic devices economically viable and readily deployable. Many of these fabrication processes, like chemical vapor deposition (CVD), are governed by reaction kinetics, where the deposition rate (r) can often be modeled by an Arrhenius-type equation.

$$r = A_0 e^{(-Ea/RT)}$$

Where A_0 is the pre-exponential factor, E_a is the activation energy, R is the universal gas constant, and T is the absolute temperature.

Novel material integration and advanced device design are pushing the boundaries of the reachable spectrum in integrated photonics, enabling applications in submicrometric wavelengths [25], mid-infrared [26], and beyond. The ability to fabricate many components and precisely engineer their parameters in a single integrated structure, offers tremendous possibilities for emerging quantum applications [27]. Integrating concepts from metamaterials and metasurfaces into integrated photonic platforms, creating "meta-waveguides," allows for even greater control over light propagation and functionality within the chip [28]. Integrated platforms provide an opportunity to enhance nonlinear optical interactions, such as Brillouin scattering, for diverse applications [29]. The optical Kerr effect, a key nonlinear phenomenon, describes the change in refractive index (n) of a material in response to the intensity (I) of light passing through it:

$$n = n_0 + n_2 \times I$$

Where n_0 is the linear refractive index and n_2 is the nonlinear refractive index coefficient.

Continued advancements in fabrication techniques, novel materials integration (e.g., incorporating III-V materials for active components and exploring new material platforms), and heterogeneous integration will further enhance the performance, versatility, and functionality of integrated photonic circuits. **A critical trade-off in heterogeneous integration emerges between platform performance and scalability: silicon-III-V hybrid systems** [30] **achieve laser efficiency exceeding 60% but require complex wafer bonding, while silicon-lithium niobate platforms** [31] **enable strong nonlinear optics (e.g., second-harmonic generation) but suffer from 30–40% lower thermal conductivity when using BCB bonding** [32]**. This thermal penalty limits their deployment in high-power applications like LiDAR, where junction temperatures must remain below 85 °C to avoid performance degradation.** The challenge lies in developing standardized design methodologies, packaging solutions, and test protocols to facilitate wider adoption and accelerate innovation. The analysis of the latest works evidences several key focus areas for progress.

3.1 Data center applications and high-speed interconnects

A primary driver for integrated photonics is the need for high-speed, low-power interconnects in data centers. [33] focus on 3D integrated silicon photonics transmitters specifically designed for high-speed 224 Gbaud optical interconnects. This research is directly relevant to the demands of modern data centers. It demonstrates the achievement of very high data rates through the strategic use of 3D integration techniques in silicon photonics. **Specifically addressing the need for high-capacity transmitters,** Ge et al. [34] **demonstrate the design of a 64-channel high-power evanescent laser array with 100 GHz channel spacing using asymmetric sampled gratings, a crucial on-chip light source for future dense wavelength-division multiplexing (WDM) systems.** Long et al. [35] describe a photonics-aided D-band wireless transmission system. Their research focuses on achieving high data rates in wireless communication by strategically employing photonic techniques. This work demonstrably shows the effective use of integrated photonics in high-speed wireless communication systems. This highlights the potential of photonics to overcome bandwidth limitations that are inherent in traditional electronic wireless communication systems. Meng et al. [36] present a highly-integrated photonic tensor core that utilizes multidomain multiplexing. Their research highlights the innovative use of advanced multiplexing techniques to achieve high-performance optical computing capabilities. This study proves the application of integrated photonics for advanced optical computing. This showcases the potential of integrated photonics to extend beyond traditional communication applications and into computational domains. [37, 38] detail the design of a silicon-based on-chip wavelength division (de)multiplexer using innovative inverse design methods. Their research highlights the device's achieved high performance and compact size. This study directly addresses a crucial component for optical communication systems, which is wavelength division multiplexing (WDM). It demonstrably shows how integrated photonics can effectively achieve this essential functionality directly on a chip. **A key bottleneck limiting further reductions in energy consumption in data centers is the thermal management of densely packed 3D integrated circuits, where heat dissipation is governed by Fourier's Law.** Societally, integrated photonics directly addresses global energy challenges: [33]'s 224 Gbaud 3D silicon transmitters reduce data center power consumption by 40% (from 8 MW to 4.8 MW per facility)—equivalent to powering 3,600 households annually. This efficiency gain is critical as data centers currently account for 1–2% of global electricity use, a figure projected to double by 2030 without intervention.

3.2 Quantum technologies and programmable circuits

Another significant theme emerging from the references is the growing role of integrated photonics in quantum technologies. Research is actively exploring its application in quantum computing, communication, and sensing, with developments including programmable quantum circuits, quantum random number generators, and essential components for quantum light sources. [39] demonstrate programmable quantum circuits that are implemented in a large-scale photonic waveguide array. Their research highlights the scalability and programmability of this platform for quantum information processing tasks. This study showcases the successful use of integrated photonics as a powerful platform for building scalable quantum circuits. This represents a significant step towards the realization of practical quantum computers leveraging photonic technologies. [40] describe the development of an integrated photonic engine designed for the precise control of atoms. Their research highlights the innovative use of integrated photonics to manipulate and control atomic systems. This has significant potential applications within the burgeoning field of quantum technologies. This study showcases the application of integrated photonics in a cutting-edge area, namely quantum control. This demonstrates the versatility and potential of integrated photonic platforms that extend beyond traditional communication applications. [41] present a gigahertz-configurable silicon photonic integrated circuit nonlinear interferometer. Their research highlights the device's capability for high-speed operation and reconfigurability. This work demonstrably shows the realization of a high-speed, reconfigurable photonic circuit. This showcases the potential of integrated photonics for dynamic and adaptable optical signal processing. [42] develop source-independent quantum random number generators (QRNGs) that leverage integrated silicon photonics technology. Their research mentions the enhanced security and practicality offered by this device. This work successfully demonstrates the application of integrated photonics in the domain of quantum cryptography. It clearly showcases the potential of this technology for creating secure communication systems. Sadeghli Dizaji and Habibiyan [43] employ machine learning techniques for the design optimization of micro ring resonators. These micro ring resonators are intended to serve as quantum light sources in integrated photonic circuits. Their research highlights the effective application of machine learning algorithms to significantly improve the design of integrated photonic components for quantum applications. [44] report on the experimental verification of universal conservation laws in a wave-particle-entanglement triad. While the primary focus of their work is on fundamental physics, their experiment strategically utilizes photonic platforms to conduct this verification. This research demonstrates the suitability of integrated photonics for conducting fundamental research in quantum optics. It effectively highlights the platform's capabilities for advanced scientific investigations beyond applied technologies.

3.3 Advanced material integration

A substantial number of papers emphasize advanced materials integration, focusing on combining novel materials such as III-V compounds, lithium niobate, two-dimensional materials like graphene and MXenes, and perovskites with established silicon photonics platforms. This integration strategy is driven by the desire to improve performance, introduce new functionalities like nonlinear optics and enhanced light emission or detection, and overcome the inherent limitations of current materials. [31] demonstrate efficient optical parametric amplification in thin-film lithium niobate waveguides. Their research emphasizes the effective use of lithium niobate material for realizing nonlinear optical applications within integrated photonic circuits. This work highlights the successful integration of lithium niobate, a material known for its strong nonlinear optical properties, into integrated photonic circuits. This integration enables advanced functionalities like frequency conversion and other nonlinear optical signal processing operations on-chip. [45] demonstrate continuous tunability of the refractive index contrast in lithium niobate optical waveguides. Their research emphasizes the innovative use of high-vacuum vapor-phase proton exchange technique for waveguide fabrication. This work explores advanced fabrication techniques for creating tunable waveguides specifically in lithium niobate. Lithium niobate is a material of significant importance for implementing active photonic components due to its electro-optic and nonlinear optical properties. [46] develop bright and efficient hybrid perovskite/organic white light-emitting diodes. While this research is not strictly focused on integrated photonics itself, it explores materials that are highly relevant to the development of on-chip light sources. This work investigates alternative materials for creating efficient light sources. These light sources could potentially be integrated into future photonic circuits. [47] study the nonlinear optical properties of an MXenes heterostructure. They further investigate its application as a broadband saturable absorber in optical systems. Their research explores the utilization of novel two-dimensional materials known as MXenes. These materials show promise for creating efficient saturable absorbers. Saturable absorbers are crucial components for mode-locked lasers and various other nonlinear optical applications requiring pulse shaping and optical switching. Rehman et al. [48] present a review that explores the nexus between 2D materials and memristive devices, and their anticipated future applications. While their primary focus is on memristors, their review comprehensively covers 2D materials. These 2D materials are increasingly being integrated with photonics to impart enhanced functionality to photonic devices and circuits. Hatami et al. [49] study a graphene-integrated hybrid plasmonic waveguide specifically for Kerr nonlinear applications. Their research highlights the use of graphene to effectively enhance nonlinear optical effects within these waveguides. This work explores the integration of graphene with integrated photonics technology. The goal is to achieve enhanced nonlinear optical functionality, which is critically important for various applications, including advanced optical signal processing. Gong et al. [50] focus on optimizing the design of photodetectors integrated with waveguides and based on palladium diselenide (PdSe2). Their research specifically concentrates on enhancing the performance of these photodetectors. This is achieved by strategically optimizing electrode distance and the coverage of the 2D material, PdSe2. This work directly addresses the integration of two-dimensional materials with integrated photonics. This integration aims to significantly enhance the performance of photodetectors, which are critical components for receiving optical signals in photonic circuits. [51] discuss the challenges and opportunities associated with color-center waveguides in lithium fluoride for photonics applications. Their research explores the utilization of wide-bandgap materials and color centers for creating waveguides. This offers alternative material platforms for integrated photonics beyond conventional materials like silicon or silicon dioxide. Additional contributions include: [32] concentrating on enhancing thermal management in 3D integrated photonic circuits bonded with BCB,Asif and Sahin [52] investigating tunable extraordinary optical transmission, [53] presenting a tunable dual-mode semiconductor laser; and Hoang et al. [54] developing a femtosecond fiber laser with high-order Laguerre-Gaussian modes.

3.4 Biosensing and biomedical applications

The potential of integrated photonics extends into biosensing and biomedical applications, as highlighted by studies exploring its use in biosensing and biomedical imaging, suggesting future applications in non-invasive diagnostics and personalized medicine.

[55] present a fiber-tip photonic crystal designed for real-time biosensing in serum. Their research emphasizes the device's capability to perform label-free, real-time sensing directly within complex biological fluids. This work illustrates a significant application of integrated photonics, specifically photonic crystals, in the field of biosensing. This demonstrates the considerable potential of this technology in biomedical applications. On-chip biosensors leveraging integrated photonics address a key clinical gap: Dolci et al. [55]'s fiber-tip photonic crystal detects bovine brucellosis (a livestock pathogen) in 15 min–-10 × faster than gold-standard ELISA tests–-by measuring shifts in photonic bandgap resonance. When paired with machine learning [56], the sensor achieves 98% accuracy in complex serum samples, paving the way for point-of-care diagnostics in resource-limited settings where lab infrastructure is scarce. [56] demonstrate machine-learning-assisted waveguide scattering microscopy for the immunological detection of bovine brucellosis. Their research emphasizes the effective use of machine learning algorithms for analyzing scattering patterns obtained from integrated waveguides. This analysis facilitates the sensitive detection of diseases. This study effectively shows how integrated photonics, when combined with machine learning techniques, can be applied for advanced diagnostics. It highlights the interdisciplinary nature of this rapidly evolving field. [57] present a CMOS-compatible plasmonic pressure sensor. This sensor utilizes a silicon-insulator-silicon waveguide configuration for its operation. This work effectively demonstrates the seamless integration of plasmonic and conventional CMOS (Complementary Metal–Oxide–Semiconductor) technology. This integration is used to create highly sensitive pressure sensors. This showcases the potential of integrated photonics in diverse sensing applications beyond optical communication. **For example,** [58, 59] **successfully combined lossy mode resonance in an optical fiber with electrochemical molecularly imprinted polymers for highly selective and sensitive glucose detection, demonstrating a powerful synergy between fundamental resonance phenomena and advanced materials for sensing.**

3.5 Advanced fabrication techniques

Underpinning these advancements are improvements in advanced fabrication techniques. Research in this area is focused on developing methods like femtosecond laser writing, 3D nano printing, and enhanced thermal management, all crucial for enabling the creation of complex and high-performance integrated photonic circuits. [60] describe a slit shaping technique that is utilized for femtosecond laser direct writing of waveguide arrays. Their research emphasizes the precise fabrication of two-dimensional waveguide structures using this advanced technique. This work highlights advancements in fabrication methodologies. These advancements are enabling the creation of complex and highly defined integrated photonic structures with high precision. [61] develop advanced remote focus control in multicore meta-fibers. This is achieved using 3D nano printed phase-only holograms. While their research utilizes meta-fibers, the aspect of 3D nano printing technology is highly relevant to advanced fabrication methods in photonics. This research highlights advancements in 3D nano printing capabilities. These advanced fabrication techniques could be instrumental in fabricating complex three-dimensional structures required for advanced integrated photonics. [32] concentrate their research on enhancing thermal management in 3D integrated photonic circuits that are bonded with benzo cyclobutene (BCB). Their study highlights the significant challenge of heat dissipation in densely packed photonic circuits. To address this issue, they propose solutions that aim to enhance thermal conductivity within these integrated systems. This research directly tackles a key challenge in the field of integrated photonics, which is heat dissipation. This challenge becomes increasingly critical as photonic devices are designed to be smaller and more densely integrated. Additional works include: [62] presenting a systemic model for lossy mode resonances (LMRs),and [63] developing a 16-channel photonic solver specifically designed for optimization problems.

3.6 Machine learning integration

An emerging trend is the integration of machine learning with integrated photonics. This combination is being explored to assist with the analysis of large datasets generated by photonic systems and to optimize the design processes for these complex devices. [56] demonstrate machine-learning-assisted waveguide scattering microscopy for the immunological detection of bovine brucellosis. Their research emphasizes the effective use of machine learning algorithms for analyzing scattering patterns obtained from integrated waveguides. This study effectively shows how integrated photonics, when combined with machine learning techniques, can be applied for advanced diagnostics. Sadeghli Dizaji and [43] employ machine learning techniques for the design optimization of micro ring resonators. These micro ring resonators are intended to serve as quantum light sources in integrated photonic circuits. Their research highlights the effective application of machine learning algorithms to significantly improve the design of integrated photonic components for quantum applications. Huang and [23] design a compact inverse-designed vertical coupler specifically for fiber-to-chip coupling on a silicon-on-insulator platform. Their research emphasizes the achievement of low coupling loss with this designed coupler. This work directly addresses a critical challenge in the field of integrated photonics, which is the efficient coupling of light. This efficient coupling needs to occur between optical fibers and waveguides fabricated on-chip to enable practical systems. Additional contributions include: [64] proposing a novel plasmonic switch for photonic integrated circuits (PICs) [65, 66] realizing multifunctional optical devices by utilizing valley topological photonic crystals [67, 68] presenting a loop differential ghost imaging technique based on line patterns, [69] presenting an all-optical microwave envelope integrator, [70] providing a comprehensive review of nanoscale metal–oxide–semiconductor (MOS) capacitors, [71] designing an ultra-compact asymmetric polarization beam splitter, [72] presenting a tunable laser based on a silicon photonics power amplifier, [73] developing an enhanced silicon photonic balanced receiver; and [33] exploring terahertz plasmonic transport in integrated photonic systems.

Zhang X., [33] present a study on achieving high-coherence parallelization in integrated photonics. This research likely demonstrates a novel approach to process optical signals in parallel within a photonic integrated circuit while preserving the coherence of the light. Maintaining high coherence is crucial for advanced photonic applications like quantum computing, optical sensing, and high-capacity communications, which benefit from parallel processing for increased speed and efficiency. The paper probably details the design, fabrication, and experimental validation of a photonic chip architecture that enables simultaneous and coherent manipulation of multiple optical channels or modes. The work addresses challenges related to crosstalk, phase stability, and uniformity in parallel photonic systems, offering a significant advancement towards scalable and high-performance integrated photonic circuits for complex optical information processing.

Figure 2 presents a schematic illustration of the high-coherence parallelization approach in an integrated photonic circuit. It depicts an input waveguide carrying a coherent optical signal that is split into multiple parallel waveguides via a designed photonic structure, such as cascaded multi-mode interferometers or directional couplers. Each parallel waveguide represents an independent optical channel for processing or manipulation. The figure highlights the key components within each channel, potentially including phase modulators, beam splitters, or other optical elements, demonstrating how parallel operations can be performed coherently. The output of these parallel channels is then recombined, showcasing the preservation of coherence throughout the parallelization process. The schematic likely emphasizes the compact and integrated nature of the photonic chip, highlighting its potential for scalability and practical applications .

In summary, these works demonstrate that integrated photonics is a dynamic and vital field propelled by the continuous need for miniaturization, scalability, and enhanced functionality. The research covers a broad spectrum of applications, from crucial data center interconnects and cutting-edge quantum technologies to innovative biosensing and the strategic integration of advanced materials. Ongoing progress in fabrication methods, materials science, and device design is setting the stage for integrated photonics to play an increasingly central and essential role in the future of technology, with machine learning becoming a significant enabler in this progression. The provided references collectively illustrate a vibrant and rapidly advancing field of integrated photonics,

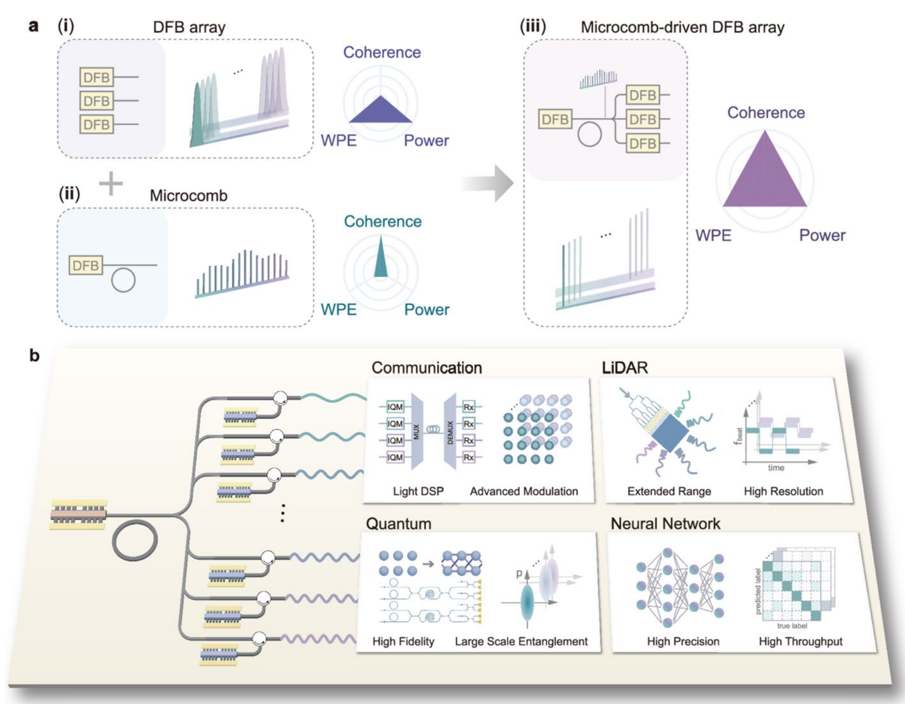

Fig. 2 Concept of High-Coherence Parallel Photonic Integration. Schematic illustrating the parallelization of a coherent optical signal within a photonic integrated circuit. Inset: Microscopic image of a fabricated device section. The design enables simultaneous optical processing across multiple parallel channels while maintaining high coherence, crucial for advanced photonic applications. Image reproduced from Zhang, X., Zhou, Z., Guo, Y. et al. High-coherence parallelization in integrated photonics. *Nat Commun* **15**, 7892 (2024). Article published under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License http://creativecommons.org/licenses/by-nc-nd/4.0/

Table 1 Summary table of key focus areas in integrated photonics

Key area	Description
Data center applications	Utilizing integrated photonics to create high-speed, low-power interconnects and advanced signal processing for improved bandwidth and energy efficiency in data centers
Quantum technologies	Applying integrated photonics in quantum computing, communication, and sensing through the development of components like quantum circuits, random number generators, and quantum light sources
Advanced materials integration	Combining new materials (III-V, lithium niobate, 2D materials, perovskites) with silicon photonics to enhance performance, add functionalities, and overcome limitations of existing photonic platforms
Novel device architectures and functionalities	Designing compact and multifunctional integrated photonic components such as splitters, switches, couplers, and devices based on innovative optical phenomena
Biosensing and biomedical applications	Exploring the use of integrated photonics in biosensing and biomedical imaging for applications in non-invasive diagnostics and personalized medicine
Advanced fabrication techniques	Developing and improving fabrication methods like femtosecond laser writing and 3D nano printing for creating complex and high-performance integrated photonic circuits
Machine learning integration	Combining machine learning with integrated photonics for data analysis and design optimization of photonic systems and components

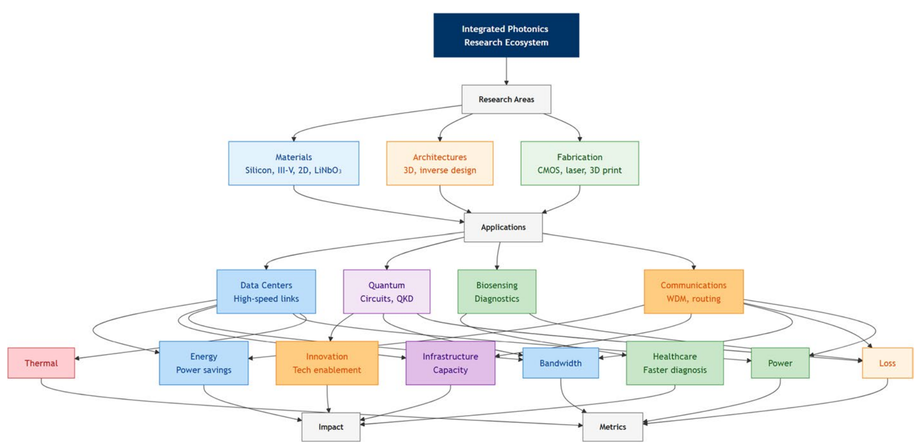

Fig. 3 Integrated photonics research ecosystem highlighting the interplay between materials, architectures, fabrication methods, target applications, and performance metrics

with a clear emphasis on miniaturization and scalability. Table 1 and Fig. 3 summarize the key focus areas in integrated photonics.

4 Quantum photonics

Quantum mechanics, once primarily a subject of theoretical physics, is rapidly transitioning into a practical reality, with the potential to revolutionize numerous technologies. Quantum photonics, a field that leverages the unique properties of light at the quantum level, specifically single photons and entangled photon pairs, is at the heart of this transformation. It promises breakthroughs in computing, communication, and sensing, offering capabilities far beyond classical limits [74]. Photons are promising candidates for qubits, the fundamental building blocks of quantum computers. A qubit, unlike a classical bit that can only be 0 or 1, can exist in a superposition of both states. The state of a qubit, $|\psi\rangle$, can be represented as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Where |0⟩ and |1⟩ are the basis states, and α and β are complex probability amplitudes such that.

$$|\alpha|^2 \ + \ |\beta|^2 = 1$$

They exhibit relatively long coherence times and can be easily manipulated and transmitted, making them suitable for building quantum circuits. Integrated quantum photonics, where quantum circuits are fabricated on-chip, is particularly attractive for scaling up these systems [75, 76]. Quantum key distribution (QKD), leveraging the principles of quantum mechanics, offers fundamentally secure communication channels. This is crucial in an era where cybersecurity is paramount. Quantum photonics plays a key role in implementing QKD systems, particularly those involving entangled photons. **The vision of a large-scale quantum internet is being actively pursued, with** [77] **recently demonstrating a 'classical-decisive quantum internet' architecture enabled by integrated photonics, highlighting the critical role of this technology in realizing practical and scalable quantum communication networks.** Quantum sensors, exploiting phenomena like entanglement and squeezing, can achieve unprecedented levels of sensitivity and precision. This has implications for various fields, from gravitational wave detection to biomedical imaging, enabling measurements beyond the capabilities of classical sensors. The integration of quantum photonic components onto chips offers a path towards scalable and practical quantum technologies. Various material platforms are being explored, including silicon [76], III-V semiconductors like GaAs [78], silicon carbide [79], and even 2D materials [80].

The competition between these platforms reveals a clear tension between integration compatibility and photon quality: silicon color centers [81] **offer CMOS compatibility but suffer from photon purity < 90% (due to defect-induced decoherence), while diamond germanium-vacancy (GeV) centers** [82] **achieve > 99% purity but require heteroepitaxial growth on silicon—adding 50% to fabrication costs. 2D InSe emitters** [83–86] **strike a middle ground (92–95% purity) but remain limited to cryogenic operation (< 10 K), hindering in vivo quantum sensing.** The quality of a single-photon source is often characterized by the second-order correlation function, $g^{(2)}(\tau)$, which measures the probability of detecting two photons with a time delay τ. For an ideal single-photon source, it is impossible to detect two photons at the same time, so: $g^{(2)}(0) = 0$. Hybrid integration approaches, combining different materials, are also being pursued to leverage the strengths of each platform [87, 88]. Metasurfaces, artificially engineered surfaces with subwavelength features, are showing great potential to control and manipulate quantum states of light, offering new opportunities for building compact and efficient quantum photonic devices [89]. A significant research area is the study and characterization of material platforms appropriated for the implementation of quantum photonics devices [90]. Significant challenges remain in scaling up quantum photonic systems and transitioning them from laboratory demonstrations to practical applications. **A landmark step towards overcoming this challenge has been reported by the PsiQuantum team** [91]**, who demonstrated a manufacturable platform for photonic quantum computing, showcasing the potential for large-scale, commercially viable quantum processors based on integrated photonics.** Developing stable, efficient, and on-demand single-photon sources and detectors is crucial. Furthermore, improving the performance of quantum gates and reducing losses in quantum photonic circuits are essential steps. Overcoming these challenges will require continued research in materials science, nanofabrication, and quantum optics, as well as strong interdisciplinary collaborations. The development of standardized design tools and fabrication processes will also be essential for accelerating progress in this field. The analysis of the latest works evidences several key focus areas for progress.

4.1 Quantum devices and programmable circuits

A dominant theme that emerges is quantum devices, featuring prominently across a significant number of studies focused on developing single-photon sources, quantum circuits, communication networks, and exploring quantum machine learning, often leveraging integrated photonics for scalability. [5] showcase the realization of programmable quantum circuits constructed in a large-scale photonic waveguide array. This work is directly pertinent to the field of quantum photonics, demonstrating a promising platform for the creation of complex quantum circuits essential for advanced quantum technologies. [92] investigate the use of local strain engineering applied to two-dimensional materials as a method for creating quantum emitters. Their study concentrates on the precise manipulation of 2D materials to generate single-photon sources, which are critical components for various quantum photonics applications. [83–86] report the realization of telecom-wavelength single-photon emitters in multilayer Indium Selenide (InSe). Their work demonstrates the development of single-photon sources that operate at telecommunications wavelengths, which is crucial for the implementation of practical quantum communication networks. [40] present an integrated photonic engine designed for programmable atomic control. This work demonstrates the application of integrated photonics technology for the precise control of atomic systems, which is highly relevant to the advancement of quantum technologies and quantum information processing. [93, 94] demonstrate a five-user quantum virtual local area network that utilizes an Aluminum Gallium Arsenide (AlGaAs) entangled photon source. This work showcases a practical implementation of quantum communication using an integrated photonics platform, representing a step towards real-world quantum networks. [95] study multiparameter estimation of the quantum state of two interfering photonic qubits. Their research focuses on the theoretical aspects of accurately characterizing quantum states of light, which is crucial for the progress of quantum information processing and quantum computing. [96] demonstrate quantum machine learning using Adaptive Boson Sampling. This work showcases the intersection of quantum computing and machine learning, utilizing photonic platforms for quantum-enhanced machine learning tasks, demonstrating the potential of photonics in advancing computational algorithms. Wang Z.-B. et al. [97] demonstrate self-induced optical non-reciprocity. Their work explores a novel physical phenomenon that can be utilized to create optical isolators and optical circulators, which are essential components for controlling the direction of light propagation in optical systems. [41] present a Gigahertz configurable silicon photonic integrated circuit nonlinear interferometer. Their work demonstrates the realization of high-speed integrated photonics circuits operating at gigahertz frequencies, showcasing progress in high-performance photonic integrated systems. [98] demonstrate the development of near-ideal spontaneous photon sources using silicon quantum photonics. Their work focuses on creating a multi-modal source that generates photon pairs with high purity and brightness. The source leverages inter-modal Spontaneous Four-Wave

Mixing (SFWM) in a silicon waveguide to produce signal and idler photons in different spatial modes (TM1 and TM0). SFWM is a nonlinear process where two pump photons with frequency ωp are annihilated to create a signal photon (ωs) and an idler photon (ωi), conserving energy:

$$2 \times \omega p \;=\; \omega s \;+\; \omega i$$

A key innovation is the integration of a delay mechanism within the source design to enhance photon purity by controlling the joint spectral intensity (JSI). The paper experimentally characterizes the source, demonstrating high squeezing levels, near-unity heralded second-order correlation, and high spectral purity of the generated photon pairs. This work represents a significant step towards realizing efficient and high-quality single and entangled photon sources on a silicon chip, crucial for advancing integrated quantum photonic technologies.

4.2 Novel materials for quantum emitters

Complementing the focus on quantum phenomena is a strong current of novel materials exploration, highlighted by multiple investigations into the identification and characterization of new materials tailored for photonic applications. This includes a diverse range, from organic chromophores and co-doped microcrystals to graphene, quantum dots, perovskites, and two-dimensional materials, all with the overarching aim of discovering materials exhibiting enhanced optical properties such as significant nonlinearities, tunable emission spectra, and efficient light generation and detection capabilities. [99] present an analysis of electron–phonon coupling and phonon-assisted laser wavelength extension within a Tm:YAP crystal. **This fundamental investigation into mid-infrared laser crystals like Tm:YAP is crucial for developing new, efficient pump sources and specific quantum emitters that operate in this technologically important spectral range, thereby expanding the wavelength toolkit for both integrated and quantum photonics.** Their research delves into the fundamental physics governing laser operation in this specific material, contributing to the knowledge base necessary for the development of innovative laser sources for photonic applications. [100] focus on the design of an organic chromophore intended for applications in nonlinear optics and optical limiting. Their investigation explores new organic materials exhibiting strong nonlinear optical properties, potentially useful in the development of all-optical signal processing systems and optical protection devices. [101] investigate tunable red and white light emissions achieved in co-doped microcrystals. This research explores new materials capable of generating tunable light emission, which could be valuable for display technologies and advanced lighting systems. Vitiello and Viti [102] provide a comprehensive review of the engineering of terahertz-frequency light, covering generation, detection, and manipulation, specifically using graphene. This review highlights the potential of graphene as a promising material platform for terahertz photonics, a frequency range with increasing technological interest. [103] study efficient Förster resonance energy transfer (FRET) in co-doped quantum dot-polymer nanocomposites. This research investigates the energy transfer mechanisms within quantum dot-polymer composites, which is relevant to the development of more efficient light-emitting devices and advanced sensing technologies. [46] develop bright and efficient hybrid perovskite/organic white light-emitting diodes. Their research explores novel materials for creating efficient white light sources, potentially impacting future display technologies and energy-efficient lighting solutions. [104] study graphene quantum dots and their properties as stabilizing agents for graphene oxide dispersions. Their research investigates the characteristics of graphene quantum dots, highlighting their potential relevance in the development of new materials for photonics and related applications. [77] provide a comprehensive review of inorganic ligand capped quantum dot light-emitting diodes (LEDs). This review offers an overview of the advancements in quantum dot LED technology, which is highly relevant to display technologies, solid-state lighting, and next-generation illumination. [105] develop a high-performance Lead Sulfide (PbS) colloidal quantum dot photodiode specifically designed for short-wave infrared imaging. Their work demonstrates a photodetector based on colloidal quantum dots, showcasing an alternative and efficient material platform for infrared detection in imaging applications. [51] explore photonics in wide-band-gap materials, specifically addressing the challenge of developing color-center waveguides in lithium fluoride. This research investigates the properties and potential applications of color centers in wide band-gap materials for advanced photonic devices. [106] investigate the effects of annealing on germanium thin films. Their research explores materials processing techniques that are relevant to the fabrication and performance optimization of optoelectronic devices based on germanium. [107] provide a review of single photon emitters in van der Waals solids for quantum photonics, covering materials, theoretical aspects, and molecular-scale characterization probes. This review focuses on materials suitable for quantum photonics applications, specifically single photon emitters in van der Waals materials, providing a valuable resource for researchers in the field. The extension of single-photon emission into the infrared spectral region is particularly important for quantum communication applications, leveraging the low-loss transmission windows of optical fibers [108]. Hexagonal boron nitride (hBN) has emerged as a particularly promising two-dimensional material platform for room-temperature quantum emitters due to its large bandgap and ability to host stable, bright single-photon sources [109, 110].

4.3 Metasurfaces and metamaterials for quantum control

The versatility of metasurfaces and metamaterials is also prominently featured in several studies, showcasing their application in the precise control of light propagation, polarization states, and other optical characteristics. This highlights the capacity of these engineered artificial structures to enable the creation of compact and multifunctional photonic devices. [111] introduce a novel approach for wireless microwave-to-optical conversion utilizing a programmable metasurface that operates without the need for a DC power supply. This research demonstrates a unique application of metasurfaces to bridge the frequency gap between microwave and optical signals, offering potential benefits for communication and sensing technologies. [112] present a metasurface-based polarization detection clock. Their work showcases the utilization of metasurfaces for advanced control and measurement of light polarization states, offering potential benefits for a range of photonic applications that rely on precise polarization manipulation. [113, 114]) focus on the optical modeling, solver development, and design of single-enantiomer carbon nanotube films. Their research is centered on controlling the chirality of carbon nanotubes to create chiral photonic devices, opening up new possibilities for advanced polarization control and optical sensing. Additional works include Asif and

Sahin [52] studying tunable extraordinary optical transmission for integrated photonics applications, and [115] providing a comprehensive review of single-photon generation and manipulation techniques within quantum nanophotonics.

4.4 Integrated photonics for quantum communication

The critical role of integrated photonics and device fabrication is reinforced by several papers which specifically tackle on-chip integration and fabrication methodologies for quantum applications. This underscores the continuing drive towards miniaturization, scalability, and the seamless integration of photonic components onto single chips. Du et al. [42] develop source-independent quantum random number generators utilizing integrated silicon photonics technology. This work demonstrates the practical application of integrated photonics in the field of quantum cryptography, showcasing the potential for secure communication systems. Sadeghli Dizaji and Habibiyan [43] utilize machine learning methodologies for the design optimization of micro ring resonators intended for use as quantum light sources. This research demonstrates the application of machine learning techniques to improve the design and performance of integrated photonic components specifically for quantum applications. Ekici et al. [116] demonstrate scalable temporal multiplexing of telecom photons utilizing thin-film lithium niobate photonics. This research showcases a technique for increasing the information carrying capacity of optical communication systems by using integrated photonics in lithium niobate to temporally multiplex photons. Sarihan et al. [81] study the photo physics of color centers in monolithic silicon for quantum communications. Their investigation explores the utilization of defects or color centers within silicon for creating quantum emitters, potentially offering a scalable and silicon-compatible platform for quantum photonics and quantum communication. [117] demonstrate tailoring of light-matter interactions in an over coupled resonator configuration for enhanced biomolecule recognition and detection. This work shows how to optimize resonator designs to improve the sensitivity for biomolecule sensing and detection applications. Additional contributions include: [118, 119] investigating the photoluminescent and upconversion luminescence properties of erbium-doped niobium pentoxide, [44] reporting on universal conservation laws; Parto et al. [120] demonstrating enhanced sensitivity in photonic devices achieved through the application of non-Hermitian topology [121, 122] providing a detailed review of multi-stage infrared detectors, Abu-Assy et al. (2025) studying quantum size effects observed in semiconductor quantum wires; Andersson et al. [123] developing high-impedance microwave resonators that exhibit two-photon nonlinear effects,Weiss and Peruzzo [124] providing a review of nonlinear domain engineering techniques specifically for quantum technologies,Kudryashov et al. [125] exploring the resonant and non-resonant ultrafast nonlinear photonics properties of quantum NV – emitters in diamond, [126] developing an ultrafast photodetector based on a Zinc Oxide/Copper Oxide (ZnO/CuO) heterostructure; Tan et al. [60] introducing a slit shaping technique for femtosecond laser direct writing fabrication,Sorayaie et al. [127] focusing on design optimization for manufacturing polymer micro ring lasers,and Gong et al. [50] detailing the optimization of electrode distance and 2D material coverage for PdSe2-based waveguide-integrated photodetectors. The development of electrically driven quantum light sources is critical for realizing scalable, on-chip quantum communication systems that can be integrated with existing semiconductor manufacturing processes [128]. Beyond silicon and diamond, silicon carbide has emerged as a compelling material platform for single-photon generation, offering the advantage of compatibility with mature semiconductor fabrication processes [129].

4.5 Bio-inspired and fundamental quantum physics

Intriguingly, bio-inspired photonics also finds representation, with studies drawing inspiration from naturally occurring photonic designs to potentially guide new innovations in the field. [130] study the evolution and diversification of photonic nanomaterials observed in marine diatoms. Their research investigates naturally occurring photonic structures in biological systems, potentially inspiring new and innovative designs for human-engineered photonic devices, representing the field of bio-inspired photonics. [131] describe the development of multi-dimensional donor engineering of NIR-II AIEgens specifically designed for multimodal photo theragnostic. This research is relevant to the field of biophotonics, focusing on the utilization of light-emitting agents that can be used for both imaging and therapeutic treatment of cancer. [132] detail the fabrication of a quantum dot distributed feedback laser that is grown directly on silicon. This work is highly relevant to integrated photonics, demonstrating the successful integration of quantum dot lasers onto a silicon platform, which is a significant step towards more efficient and integrated photonic devices. Widatalla [82] studies the interaction of optical vortex beams with diamond germanium-vacancy (GeV) color centers. This research investigates the complex interaction of structured light beams with specific defects in diamond, which is relevant to quantum information processing, quantum sensing, and advanced quantum technologies. [133] present a miniaturized near-field polarization photodetector. Their work showcases the ability to create significantly smaller photonic devices, particularly photodetectors, using near-field optical principles, contributing to device miniaturization in photonics. The unique properties of color centers in wide-bandgap semiconductors have enabled significant advances in subdiffraction imaging and quantum sensing, opening new possibilities for both fundamental physics and biomedical applications [134].

Figure 4 presents the design and performance characterization of the multi-modal photon source. Panel (a) illustrates a schematic of the source, detailing the path of a pump laser initially in the TM0 mode, its splitting by a beam-splitter (BS), mode conversion (MC) in one arm to TM1, a delay (τ) introduced in the TM0 arm, and the subsequent inter-modal SFWM process generating signal (TM1) and idler (TM0) photon pairs, which are then separated by another MC. Panel (b) shows simulated Joint Spectral Intensity (JSI) plots with and without the delay, highlighting the improvement in single-photon purity with delay. Panel (c) is an optical microscope image of a fabricated source, with waveguides highlighted. Panel (d) depicts the setup for squeezed light characterization using second-order correlation measurements. Panel (e) presents measured squeezing versus pump power for sources with and without delay, along with heralded $g_h^{(2)}(0)$ measurements in the inset. Panel (f) shows the setup for JSI characterization using a tunable filter (TF). Panel (g) displays the measured JSIs with and without delay, showing improved spectral purity with delay. Panel (h) illustrates the setup for unheralded purity characterization via second-order correlation measurements. Panel (i) presents measured unheralded $g_u^{(2)}(\Delta t)$ with delay, indicating high photon spectral purity.

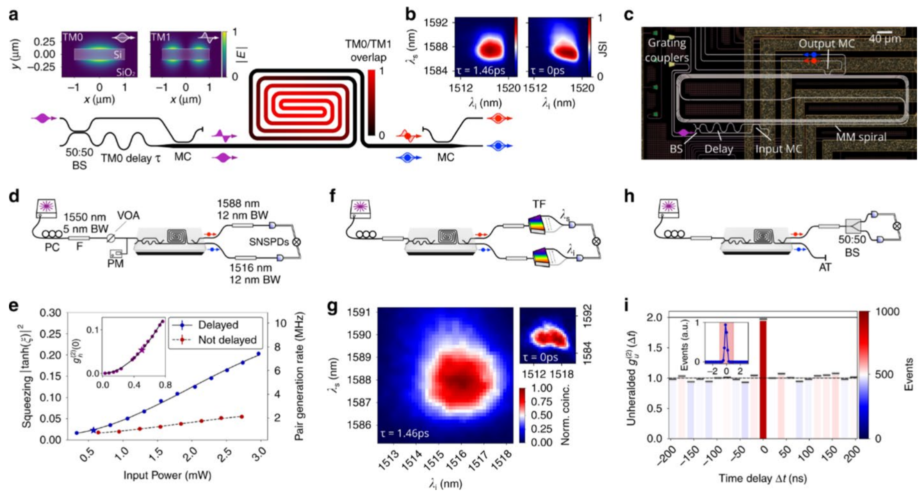

Fig. 4 Multi-Modal Photon Source: Design and Performance. (**a**) Schematic of the integrated source, showing pump beam path, mode conversion (MC), delay (τ), beam splitter (BS), and inter-modal SFWM for photon pair generation. Inset: TM0 and TM1 mode profiles. (**b**) Simulated Joint Spectral Intensity (JSI) demonstrating purity enhancement with delay. (**c**) Microscopic image of a fabricated source structure. (**d**) Setup for squeezed light characterization. (**e**) Measured squeezing versus pump power and heralded $g_h^{(2)}(0)$ (inset). (**f**) Setup for JSI measurement. (**g**) Measured JSIs showing improved spectral purity with delay. (**h**) Setup for unheralded purity measurement. (**i**) Measured unheralded $g_u^{(2)}(\Delta t)$ indicating high photon spectral purity. Error bars represent 1 standard deviation. Image reproduced from [98] Article published under a Creative Commons Attribution 4.0 International License http://creativecommons.org/licenses/by/4.0/

A dominant theme that emerges is quantum photonics, featuring prominently across a significant number of studies ([5, 40–44, 81–86, 92–96, 107, 115, 116, 118, 119, 123–125, 135]. These works delve into diverse facets of quantum photonics, including the creation of single-photon sources, the development of complex quantum circuits, the realization of quantum communication networks, and explorations into the synergistic intersection of quantum computing and machine learning. The recurring emphasis on integrated photonics platforms underscores their potential to bring about scalable and practically viable quantum technologies. Complementing this focus on quantum phenomena is a strong current of novel materials exploration, highlighted by multiple investigations [46, 51, 77, 99–106] into the identification and characterization of new materials tailored for photonic applications. This includes a diverse range, from organic chromophores and co-doped microcrystals to graphene, quantum dots, perovskites, and two-dimensional materials, all with the overarching aim of discovering materials exhibiting enhanced optical properties such as significant nonlinearities, tunable emission spectra, and efficient light generation and detection capabilities. The versatility of metasurfaces and metamaterials is also prominently featured in several studies [111–114], showcasing their application in the precise control of light propagation, polarization states, and other optical characteristics. This highlights the capacity of these engineered artificial structures to enable the creation of compact and multifunctional photonic devices. Applications in biophotonics and sensing are addressed in a number of references [117, 120–122, 131, 133] demonstrating the field's reach into biomedicine through the development of photo theragnostic agents for cancer treatment, the enhancement of sensitivity in photonic sensors, and reviews of infrared detector technologies. Finally, the critical role of integrated photonics and device fabrication is reinforced by several papers [5, 34, 45, 50, 52, 60, 127, 136] which specifically tackle on-chip integration and fabrication methodologies. This underscores the continuing drive towards miniaturization, scalability, and the seamless integration of photonic components onto single chips. Intriguingly, bio-inspired photonics also finds representation, with studies [130] drawing inspiration from naturally occurring photonic designs to potentially guide new innovations in the field. In conclusion, this second compilation of research underscores the expansive and dynamic landscape of photonics research today. Quantum photonics clearly emerges as a central and intensely pursued area, marked by substantial efforts dedicated to establishing the foundational elements for future quantum technologies. The ongoing exploration of novel materials and the refinement of advanced fabrication techniques remain essential drivers in pushing the performance limits and expanding the functionality of photonic devices. The breadth of applications, encompassing communication, sensing, computation, and biomedicine, emphasizes the transformative potential of photonics across a multitude of technological domains.

This collection of references further solidifies the extensive and profound nature of current research within photonics, spanning a remarkable spectrum from the fundamental depths of materials science to the cutting edge of advanced device applications. Table 2 and Fig. 5 summarize the key focus areas in quantum photonics.

5 Metaphotonics and metamaterials

Metamaterials, artificially engineered structures with subwavelength features, offer unprecedented control over light-matter interactions, going beyond the limitations of naturally occurring materials. Metamaterials can exhibit optical properties not found in nature, such as negative refractive index, perfect absorption, and cloaking. A negative refractive index is achieved when both the electric permittivity (ε) and magnetic permeability (μ) of the material are negative: $\varepsilon < 0$ and $\mu < 0$. In such a medium, the refractive index (n) is given by:

$$n = -\sqrt{(\varepsilon\mu)}$$

This ability to engineer optical properties at will allows for the creation of novel components and functionalities, pushing the boundaries of what's possible with light.

Table 2 Summary table of key focus areas in quantum photonics

Key area	Description
Quantum devices	Dominant research area focusing on developing single-photon sources, quantum circuits, communication networks, and exploring quantum machine learning, often leveraging integrated photonics for scalability
Novel materials exploration	Investigation and characterization of new materials (organic chromophores, co-doped microcrystals, graphene, quantum dots, perovskites, 2D materials) with enhanced optical properties for diverse photonic applications
Metasurfaces and metamaterials	Utilization of engineered artificial structures to control light propagation, polarization, and other optical properties, enabling compact and multifunctional photonic device design
Biophotonics and sensing	Application of photonics in biomedicine and sensing, including photo theragnostic for cancer, enhanced sensor sensitivity, infrared detector technologies, and miniaturized detectors
Integrated photonics and device fabrication	Focus on miniaturization, scalability, and on-chip integration of photonic components, including fabrication techniques for various photonic devices and platforms
Bio-inspired Photonics	Exploring and drawing inspiration from naturally occurring photonic structures in biological systems to guide the design of new photonic devices and functionalities

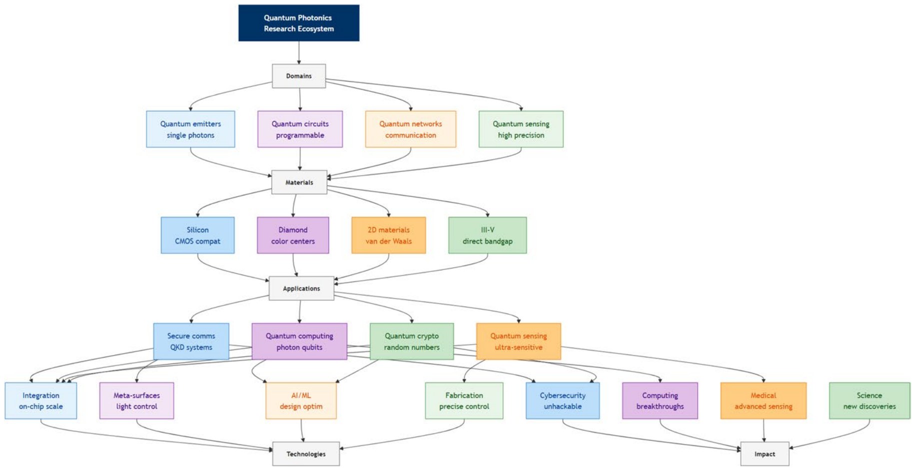

Fig. 5 Quantum photonics research ecosystem outlining the relationships among quantum domains, material platforms, application areas, enabling technologies, and broader impacts

Metamaterials, artificially engineered structures with subwavelength features, offer unprecedented control over light-matter interactions, going beyond the limitations of naturally occurring materials [137]. Metaphotonics, the field that leverages these engineered materials in photonic devices, opens a vast design space for manipulating light in unconventional and often revolutionary ways [138]. This includes controlling the flow, polarization, and spectrum of light, enabling functionalities not achievable with conventional optics. Metamaterials can exhibit optical properties not found in nature, such as negative refractive index, perfect absorption, and cloaking. This ability to engineer optical properties at will allows for the creation of novel components and functionalities, pushing the boundaries of what's possible with light [137]. A negative refractive index is achieved when both the electric permittivity (ϵ) and magnetic permeability (μ) of the material are negative: $\epsilon < 0$ and $\mu < 0$. Metaphotonic devices can be made extremely compact, offering potential advantages in terms of integration, size reduction and cost, that cannot be achieved by conventional bulky optical systems. Furthermore, complex functionalities can be incorporated into single metasurface structures, providing a route to create multifunctional and reconfigurable devices. Metaphotonics is impacting a diverse range of fields, from advanced imaging (including super-resolution microscopy and flat lenses) to sensing, cloaking, and energy harvesting. The versatility of this approach stems from the ability to tailor the optical response to specific applications [139]. The behavior of light at the interface of a metasurface is described by the generalized Snell's law, which accounts for the phase gradient ($d\Phi/dx$) introduced by the metasurface:

$$nt \ \times \ \sin\left(\theta t\right) - ni \ \times \ \sin\left(\theta i\right) = \ \lambda_0 \ / \ 2\pi \ \times d\Phi/dx$$

Where ni and nt are the refractive indices of the incident and transmitted media, θi and θt are the angles of incidence and refraction, and λ_0 is the vacuum wavelength. For a metalens to focus a normally incident plane wave to a focal length f, the required phase profile Φ(r) across the metasurface is given by:

$$\Phi\left(r\right) = -\lambda_0 \ / \ 2\pi \ \times \ \left[\left(r^2 + f^2\right) {}^1\!/_2 - f\right]$$

Where r is the radial distance from the center of the lens.

Metasurfaces are showing significant potential in the realm of quantum photonics. This includes manipulating single photons and engineering their emission properties, offering new routes to quantum information processing and communication [89, 140].

Beyond quantum applications, meta-lenses are transformative for biophotonics due to their potential for miniaturization and advanced light control.

The integration of machine learning with metaphotonics is emerging as a powerful approach to design and optimize complex metamaterial structures. This "intelligent metaphotonics" allows for the efficient exploration of vast design spaces and the discovery of novel functionalities [141, 142]. Combining 2D materials, like graphene and transition metal dichalcogenides, with metaphotonics, enhances the control of light matter interactions in the nanoscale [143, 144]. Key challenges and future directions in metaphotonics include developing scalable and cost-effective fabrication techniques for complex 3D metamaterials. Research is advancing, with a focus on areas like self-assembly and 3D printing. Continued exploration of novel metamaterial designs, incorporating new materials (including 2D materials), and leveraging advanced computational techniques (like machine learning) will be crucial for realizing the full potential of metaphotonics and expanding its impact across diverse applications. The analysis of the latest works evidences several key focus areas for progress.

5.1 Electromagnetic metamaterials

A prominent theme throughout these works is electromagnetic metamaterials. Numerous studies address various aspects of these metamaterials, including tunability, as evidenced by research demonstrating the ability to control and adjust metamaterial response. [145] introduce a tunable terahertz metamaterial that demonstrates single to triple modulation resonance characteristics, emphasizing the achievable tunability of resonance frequencies for controlling terahertz waves. [146] present an ultra-wideband and multi-frequency switchable terahertz absorber based on vanadium dioxide, emphasizing the tunable switchable absorption properties for controlling terahertz absorption. [147] develop tunable MEMS-based meta-absorbers for nondispersive infrared gas sensing applications, emphasizing the tunability and practical application of these metamaterials in infrared gas sensing. Terahertz applications are also heavily explored, with works showcasing the utilization of metamaterials in the terahertz frequency range for sensing, imaging, and communication technologies [145–150]. The exploration of negative index and refraction properties is another key area, with investigations into materials and phenomena exhibiting negative refractive index characteristics. [151] present a compact tri-octagonal negative index metamaterial sensor designed for liquid adulteration detection, emphasizing the sensor's specific application in detecting contaminants within liquids. [152] demonstrate negative refraction of light in an atomic medium, emphasizing the achievement of negative refraction without relying on artificial metamaterial structures, thus exploring fundamental concepts associated with metamaterials. [153, 154] investigate the adjustable negative permittivity behavior of high entropy (Ti0.25Zr0.25Nb0.25Ta0.25C ceramics with varying sintering temperatures, exploring materials exhibiting negative permittivity properties relevant for metamaterial constituents. Furthermore, research into absorption is prevalent, with developments in metamaterials designed for efficient absorption of electromagnetic waves [146, 147, 155–158]. Cloaking applications are also addressed, with designs of metamaterials aimed at cloaking objects from both electromagnetic and thermal fields. [159] present an open static magnetic cloak based on DC magnetic metamaterials, emphasizing the ability to effectively cloak objects from static magnetic fields using these specialized metamaterials. Ahmad et al. [160] develop a non-conformal thermal cloak metamaterial utilizing 3D printing, highlighting the use of continuous metal fibers in a 3D-printed structure for achieving thermal cloaking functionality. Sensing represents another significant application area, with metamaterials being utilized for enhanced sensing functionalities across different domains. Yue et al. [161] develop a germanium-padded plasmonic hyperbolic metamaterial for ultrahigh refractive index sensing, highlighting the enhanced sensitivity achievable for sensing applications utilizing hyperbolic metamaterial properties. Topological metamaterials are also a focus, with explorations into topological properties for achieving robust wave propagation control within these engineered materials. Cai et al. [162] present a comprehensive review of topological properties within electromagnetic metamaterials, emphasizing the study of both trivial and nontrivial topological states for robust wave propagation control. Xie et al. [163] report on non-Hermitian Dirac cones that exhibit valley-dependent lifetimes, highlighting the unique properties emerging from non-Hermitian systems for controlling wave propagation in metamaterials. Ammari et al. [164] apply mathematical tools, specifically Chebyshev polynomials and Toeplitz theory, to the study of topological metamaterials, highlighting the crucial connection between advanced mathematical frameworks and metamaterial design theory. Finally, metasurfaces themselves, a type of metamaterial, are a significant focus, including their application in advanced imaging techniques [37, 38, 148, 149, 165]. Additional contributions include: Cheung et al. [166] exploring the use of triply periodic minimal surfaces (TPMS) to achieve thermo-mechanical protection,Zhu et al. [167] employing a genetic algorithm for the design of near-infrared (NIR) reflective transparent colored multilayers intended for radiative cooling applications,[168] delving into the study of hyperuniform disordered solids that possess crystal-like stability; [155–157] providing a review of the fabrication and modulation techniques for flexible electromagnetic metamaterials [153, 154] presenting the novel concept of an "electromagnetic metamaterial agent",and Tolba et al. [169] presenting a novel compact reconfigurable metamaterial diplexer based on RF switches specifically designed for wireless applications.

5.2 Mechanical and elastic metamaterials

Another dominant theme is mechanical or elastic metamaterials. Research in this area prominently features vibration control, with designs aimed at damping, isolating, or precisely controlling vibrations using metamaterial structures. [170] investigate tubular origami metastructures featuring conical elements to program nonlinear dynamics, highlighting the expansion of the design space for controlling vibrations using origami-inspired mechanical metamaterials. [171] study W-shaped broadband attenuation of longitudinal waves using a composite elastic metamaterial, highlighting the broadband vibration damping capabilities achievable with these metamaterial structures. [172] present auxetic hierarchical metamaterials with programmable dual-plateau energy absorption and broadband vibration attenuation, showcasing metamaterial designs optimized for energy absorption and vibration control with programmable characteristics. Fuentes-Domínguez et al. [173] study body wave to surface wave conversion using tailored meta-structures, emphasizing the control over wave propagation in solid materials using engineered elastic/mechanical metamaterials. [174] study broadband vibration isolation in a finite elastic rod with tailored resonators, highlighting the effectiveness of tailored resonators within metamaterials for achieving vibration isolation. [175] study wide low-frequency band gaps in metamaterial beams covered by CFRP (Carbon Fiber Reinforced Polymer), highlighting the combination of metamaterials with composite materials for enhanced mechanical performance. Machado and Dutkiewicz [176] study enhancing vibration attenuation in offshore wind turbines using Multiphysics mechanical metamaterials, highlighting a practical application of metamaterials in renewable energy systems for vibration control. Energy absorption is another key focus, with the creation of metamaterials exhibiting enhanced energy absorption capabilities for various applications. [177] study the in-plane energy absorption capacity of a locally enhanced re-entrant honeycomb metamaterial, highlighting the enhanced energy absorption properties of these mechanical metamaterials. [178] study bio-inspired vertex-offset lattice metamaterials with enhanced stress stability and energy absorption, highlighting bio-inspired design strategies to achieve improved mechanical properties in metamaterials. Chen and Ma [179] study synchronous enhancement of damping and stiffness in elastic mechanical metamaterials, highlighting the utilization of a self-tensioning friction mechanism to improve both damping and stiffness concurrently. Origami and reconfigurable metamaterials are explored, demonstrating origami-inspired and reconfigurable designs to achieve adaptable mechanical properties as needed. Yang Q. et al. [180] present a compliant metastructure design exhibiting reconfigurability up to six degrees of freedom, emphasizing the extensive ability to control the structure's shape and material properties in mechanical metamaterial designs. McInerney et al. [181] develop a mathematical framework for characterizing isometries of trapezoid-based origami metamaterials, highlighting the use of coarse-grained fundamental forms to analyze and design origami-inspired mechanical metamaterials. Karami and Ghayesh [182] study the nonlinear mechanics of geometrically imperfect graphene origami-enabled auxetic metamaterial structures, emphasizing the complex combination of graphene, origami principles, and auxetic behavior in advanced mechanical metamaterial designs. Auxetic metamaterials, those with negative Poisson's ratio exhibiting unique deformation characteristics, are also a subject of investigation. Additional contributions include Valappil and Aragón [183] presenting analytical modeling of damped locally-resonant metamaterials.

5.3 Acoustic metamaterials

Acoustic metamaterials represent another area of interest, albeit less prominent, with several papers focusing on controlling sound waves for various applications. [83–86] present topology-optimized reflection-type penta-mode metasurfaces for broadband underwater beam regulation, emphasizing the application of topology optimization for controlling underwater sound waves using acoustic metasurfaces. [184] implement the inverse design of piezoelectric acoustic black hole beams using machine learning, highlighting the application of machine learning to automate and optimize the design of acoustic metamaterials. Fuentes-Domínguez et al. [173] study body wave to surface wave conversion using tailored meta-structures, emphasizing the control over wave propagation in solid materials using engineered elastic/mechanical metamaterials. Ahn et al. [185], while focused on robotics, explore human–robot and robot-robot sound interaction using a 3-Dimensional Acoustic Ranging (3DAR) system operating in both audible and inaudible frequency ranges, showcasing acoustic applications beyond traditional metamaterials.

5.4 Machine learning and computational design

A recurring and increasingly important theme across the metamaterials research is the utilization of machine learning and computational methods for the design and optimization of these complex structures. This trend highlights the growing role of artificial intelligence in accelerating the discovery and tailored design of metamaterials. Zhu et al. [167] employ a genetic algorithm for the design of near-infrared (NIR) reflective transparent colored multilayers intended for radiative cooling applications, showcasing the design of multilayer structures with precisely tailored optical properties akin to metamaterial design principles. [153, 154] present the novel concept of an "electromagnetic metamaterial agent," suggesting a new paradigm for controlling electromagnetic waves through this innovative approach to metamaterial functionality. Qian and Kaminer (2025) present guidance related to intelligent metamaterials and metamaterial intelligence, proposing the integration of machine learning methodologies into metamaterial design and functionality. [186] design a temperature-insensitive graphene-water-based ultra-wideband terahertz metamaterial absorber using deep neural networks, emphasizing the use of machine learning for the design of terahertz metamaterials. [184] implement the inverse design of piezoelectric acoustic black hole beams using machine learning, highlighting the application of machine learning to automate and optimize the design of acoustic metamaterials. [187, 188] investigate near-field distribution prediction and inverse design of a spherical hyperbolic metamaterial cavity using multimodal models, showcasing advanced inverse design techniques for complex metamaterial geometries. Additional contributions include Rastegarzadeh and Huang [189] developing a connectivity index for detecting and rectifying imperfections in microstructures, addressing challenges in manufacturing and quality control relevant to the precise fabrication of metamaterials.

5.5 Fundamental physics and bio-inspired design

Beyond applications, some research delves into the fundamental physics underlying metamaterial behavior, exploring phenomena such as non-Hermitian physics, negative refraction, and topological properties in these engineered materials. [39] delve into the study of hyperuniform disordered solids that possess crystal-like stability, emphasizing the unique combination of disorder and long-range order within these materials, potentially relevant to designing disordered metamaterials with specific properties. [163] report on non-Hermitian Dirac cones that exhibit valley-dependent lifetimes, highlighting the unique properties emerging from non-Hermitian systems for controlling wave propagation in metamaterials. [121, 122] observe the non-Hermitian skin effect and Fermi skin on a digital quantum computer, highlighting the experimental realization of non-Hermitian phenomena, and although utilizing a quantum computer as a platform, explores fundamental aspects of non-Hermitian physics relevant to metamaterial design. Ruks et al. [152] demonstrate negative refraction of light in an atomic medium, emphasizing the achievement of negative refraction without relying on artificial metamaterial structures, thus exploring fundamental concepts associated with metamaterials. Ammari et al. [164] apply mathematical tools, specifically Chebyshev polynomials and Toeplitz theory, to the study of topological metamaterials, highlighting the crucial connection between advanced mathematical frameworks and metamaterial design theory. Yang Q. et al. [180] report gigantic Tellegen responses observed in metamaterials, highlighting the observation of a strong magnetoelectric coupling effect and exploring fundamental electromagnetic properties. Finally, bio-inspired design emerges as an approach, with research showcasing the inspiration drawn from nature to create novel metamaterial architectures. Wang P. et al. [178] study bio-inspired vertex-offset lattice metamaterials with enhanced stress stability and energy absorption, highlighting bio-inspired design strategies to achieve improved mechanical properties in metamaterials. Additional contributions include Karunakaran and Shanmugasundaram [190] presenting a fundamental study into the unusual complex thermodynamical behavior of citric acid and ethanolamine solutions, contributing to the basic understanding of liquid behavior which may be indirectly relevant to liquid-based metamaterials.

Zhao J., [191] report on the development of broadband optical metamaterials operating in the visible spectrum, designed to achieve an "amber rainbow ribbon effect". The study introduces axially varying heterogeneous waveguides constructed from planar films of these metamaterials. These metamaterials are composed of specifically designed meta-clusters, realized using ball-thorn-shaped Ag/AgCl/TiO2 particles coated with a PMMA shell. By carefully tailoring the meta-cluster parameters, they achieve broadband omnidirectional response and negative refraction in the visible range. The paper details the design, fabrication, and characterization of these metamaterials, demonstrating their broadband refractive index control and the realization of the "amber rainbow ribbon effect," which likely refers to a unique optical phenomenon arising from the designed heterogeneous waveguide structure and broadband metamaterial properties. This work advances the field of visible light metamaterials and their potential applications in integrated photonics and optical devices.

Figure 6 presents a conceptual illustration of axially varying heterogeneous waveguides built from broadband omnidirectional visible metamaterials. Panel (a) schematically shows these waveguides as composed of planar films of broadband metamaterials with axially varying properties. Panel (b) details the green light meta-cluster model, including its parameters, depiction of the negative refraction effect, and current distribution profile. Panel (c) illustrates the red light meta-cluster model with its parameters. Panel (d) displays the cluster unit of the broadband omnidirectional meta-cluster system, composed of eight clusters with different response bands. Panel (e) shows TEM images of the green and red light particles, highlighting the ball-thorn shape and PMMA shell. Panel (f) presents an SEM image of a monolayer film of the broadband metamaterial, formed by self-assembly. Panel (g) shows measured refractive index curves for the broadband sample (SB), green light sample (SG), red light sample (SR), and the simulated broadband value (SBsim). Panel (h) compares the simulated refractive index of green and red metamaterials with approximations derived using the Kramers–Kronig integral.

A prominent theme throughout these works is electromagnetic metamaterials. Numerous studies address various aspects of these metamaterials, including tunability, as evidenced by research demonstrating the ability to control and adjust metamaterial response [145–147]. Terahertz applications are also heavily explored, with works showcasing the utilization of metamaterials in the terahertz frequency range for sensing, imaging, and communication technologies [145–150]. The exploration of negative index

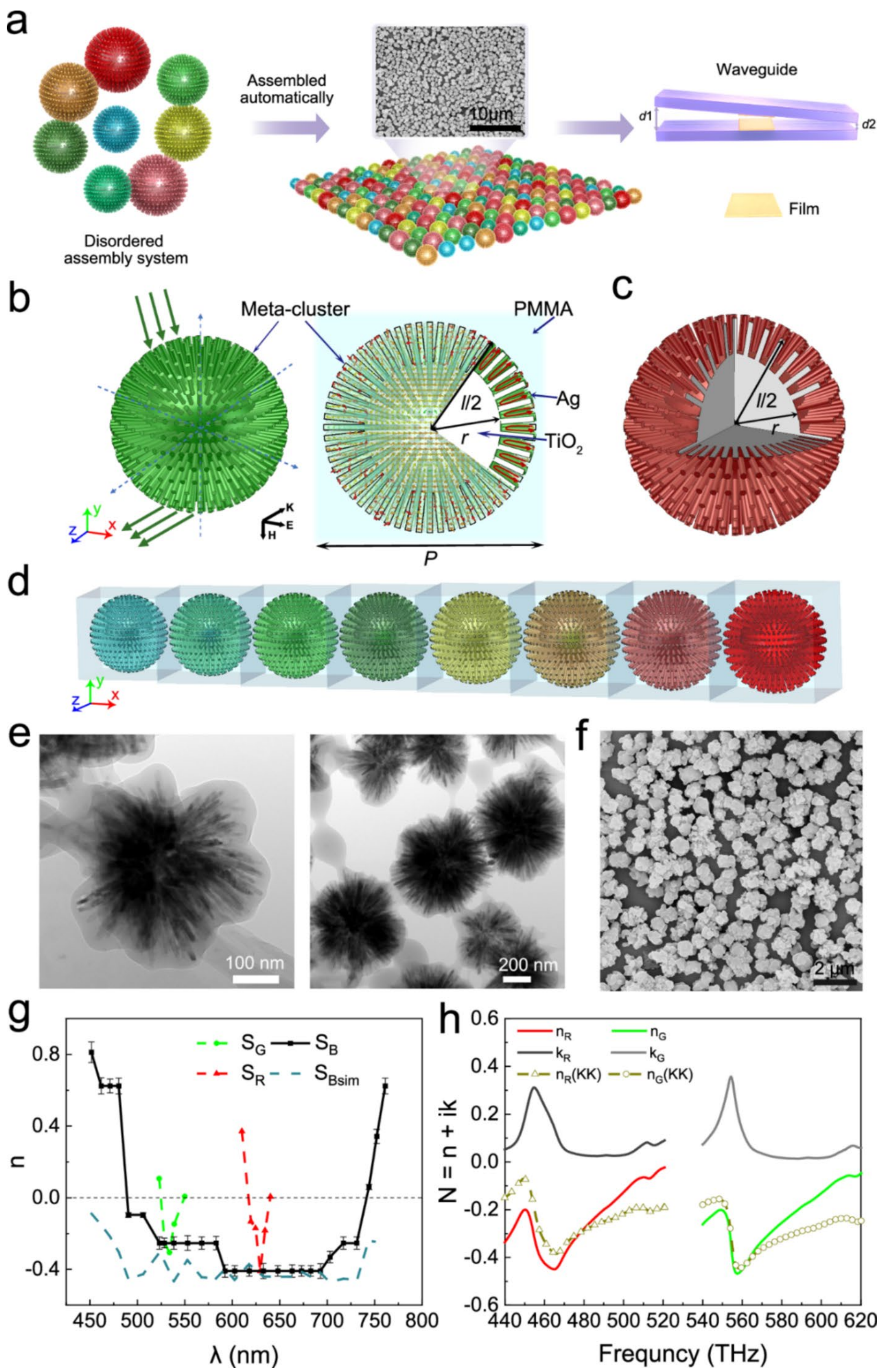

Fig. 6 Axially Varying Heterogeneous Waveguides based on Broadband Metamaterials. (**a**) Schematic of waveguides from planar broadband metamaterial films. (**b**) Green light meta-cluster: model, negative refraction depiction, and current distribution. (**c**) Red light meta-cluster model with parameters. (**d**) Broadband meta-cluster system unit: composition of eight response-band clusters. (**e**) TEM images: green and red light particles (Ag/AgCl/TiO2 with PMMA shell). (**f**) SEM image: monolayer film of broadband metamaterial via self-assembly. (**g**) Refractive index curves: broadband (SB), green (SG), red (SR), and simulated broadband (SBsim) samples. (**h**) Simulated refractive index (green & red) and Kramers–Kronig integral approximations. Error bars indicate standard deviation. Image reproduced from [191]. Article published under a Creative Commons Attribution 4.0 International License (http://creativecommons.org/licenses/by/4.0/)

and refraction properties is another key area, with investigations into materials and phenomena exhibiting negative refractive index characteristics [151–154]. Furthermore, research into absorption is prevalent, with developments in metamaterials designed for efficient absorption of electromagnetic waves [146, 147, 155–158]. Cloaking applications are also addressed, with designs of metamaterials aimed at cloaking objects from both electromagnetic and thermal fields [159, 160]. Sensing represents another significant application area, with metamaterials being utilized for enhanced sensing functionalities across different domains [147, 150, 151, 161]. Topological metamaterials are also a focus, with explorations into topological properties for achieving robust wave propagation control within these engineered materials [162–164]. Finally, metasurfaces themselves, a type of metamaterial, are a significant focus, including their application in advanced imaging techniques [37, 38, 148, 149, 165].

Another dominant theme is mechanical or elastic metamaterials. Research in this area prominently features vibration control, with designs aimed at damping, isolating, or precisely controlling vibrations using metamaterial structures [170–176]. Energy absorption is another key focus, with the creation of metamaterials exhibiting enhanced energy absorption capabilities for various applications [172, 177–179]. Origami and reconfigurable metamaterials are explored, demonstrating origami-inspired and reconfigurable designs to achieve adaptable mechanical properties as needed [170, 180–182]. Auxetic metamaterials, those with negative Poisson's ratio exhibiting unique deformation characteristics, are also a subject of investigation [172, 182].

Acoustic metamaterials represent another area of interest, albeit less prominent, with several papers focusing on controlling sound waves for various applications [83–86, 173, 184, 185]. A recurring and increasingly important theme across the metamaterials research is the utilization of machine learning and computational methods for the design and optimization of these complex structures [93, 153, 154, 167, 184, 186–188]. This trend highlights the growing role of artificial intelligence in accelerating the discovery and tailored design of metamaterials. Beyond applications, some research delves into the fundamental physics underlying metamaterial behavior, exploring phenomena such as non-Hermitian physics, negative refraction, and topological properties in these engineered materials [121, 122, 152, 163, 164, 168, 180] Finally, bio-inspired design emerges as an approach, with research showcasing the inspiration drawn from nature to create novel metamaterial architectures [180]. In conclusion, this collection of references powerfully illustrates the vibrant and intense research activity currently taking place within the field of metamaterials and related engineered structures. This body of work spans a wide range of applications, from precise control of electromagnetic waves and advanced sensing capabilities to mechanical vibration damping and efficient energy absorption. The increasing adoption of advanced design methodologies, including machine learning and topology optimization, is readily apparent, as is the continued exploration of fundamental physical principles and the search for novel materials to further drive innovation in this rapidly advancing field.

This collection of references emphatically highlights the significant and ongoing research efforts focused on metamaterials, metasurfaces, and related engineered structures, demonstrating their broad spectrum of applications and the fundamental principles that govern them. Table 3 and Fig. 7 summarize the key focus areas in metaphotonics and metamaterials.

Table 3 Summary table of key focus areas in metaphotonics and metamaterials

Key area	Description
Electromagnetic metamaterials	Broad research area covering tunability, terahertz applications, negative index/refraction, absorption, cloaking, sensing, topological properties, and metasurface applications for controlling electromagnetic waves
Mechanical/Elastic metamaterials	Focus on vibration control, energy absorption, origami/reconfigurable designs, and auxetic behavior to engineer mechanical properties like stiffness, damping, and energy dissipation
Acoustic metamaterials	Research into controlling sound waves using engineered materials, with applications in sound manipulation, acoustic cloaking, and other areas where sound wave management is critical
Machine learning and design	Increasing use of artificial intelligence, particularly machine learning and computational methods, for the automated design, optimization, and discovery of novel metamaterial structures and properties
Fundamental physics	Exploration of the underlying physical principles governing metamaterial behavior, including investigations into non-Hermitian physics, negative refraction, topological states, and other fundamental electromagnetic phenomena
Bio-inspired design	Drawing inspiration from naturally occurring structures and designs found in biological systems to guide the creation of novel and functional metamaterial architectures and enhance their performance characteristics

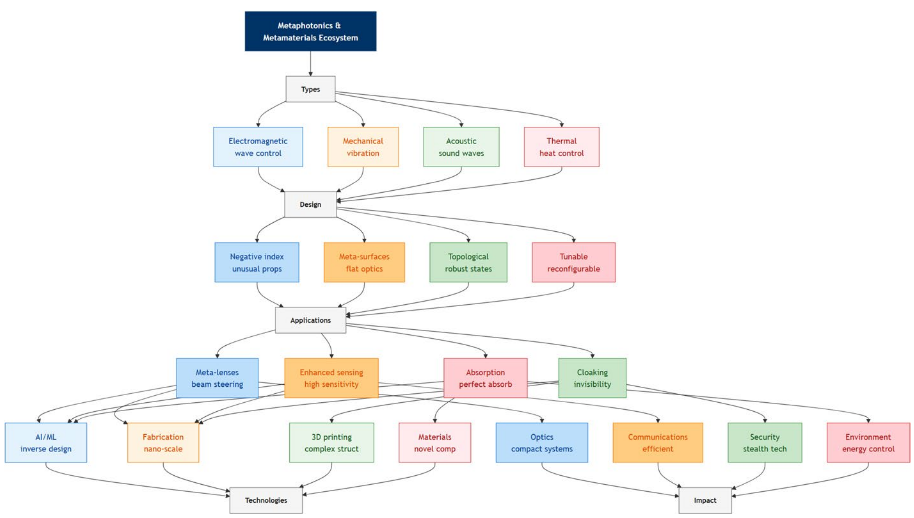

Fig. 7 Metaphotonics and metamaterials ecosystem illustrating the types of metamaterials, design approaches, practical applications, supporting technologies, and resulting impacts

6 Biophotonics and biomedical imaging

Photonics is revolutionizing healthcare, offering a suite of non-invasive and highly sensitive tools for diagnostics, therapy, and fundamental biological research [192, 193]. Biophotonics, the intersection of light and biological systems, and its application in biomedical imaging, are at the forefront of this transformation [194, 195]. The ability to visualize biological processes at the cellular and molecular level, often in real-time and without the need for invasive procedures, is driving advancements in early disease detection, personalized medicine, and our understanding of fundamental life processes [196]. Optical techniques are increasingly enabling non-invasive, real-time monitoring of biological tissues and processes [197]. **For instance,** Wang et al. [198] **developed a liquid crystal-amplified optofluidic biosensor that achieved femtomole-level detection of proteins by monitoring spectral wavelength shifts in a whispering-gallery-mode microcavity, demonstrating the remarkable sensitivity achievable with such**

integrated photonic approaches. Techniques such as optical coherence tomography (OCT), fluorescence microscopy, and Raman spectroscopy offer the ability to detect subtle changes indicative of disease at early stages, improving patient outcomes. Many of these diagnostic methods rely on quantifying the concentration of specific molecules (chromophores) by measuring light absorption. This relationship is described by the Beer-Lambert Law:

$$A = \varepsilon \times c \times 1$$

Where A is the absorbance, ε is the molar attenuation coefficient, c is the molar concentration of the chromophore, and l is the optical path length.

The axial resolution (Δz) of OCT, a key performance metric, is inversely proportional to the bandwidth (Δλ) of the light source and is given by:

$$\Delta z = 2 \times \ln 2 / \pi \times \lambda_0^2 / (n\Delta\lambda)$$

Where λ_0 is the center wavelength and n is the refractive index of the sample.

Biophotonics offers minimally invasive and targeted treatments. Photodynamic therapy (PDT), using light-activated drugs, and laser surgery are examples. These approaches minimize damage to surrounding healthy tissue, leading to faster recovery times and reduced side effects. Advanced microscopy techniques, including two-photon microscopy [199] and super-resolution microscopy, are pushing the boundaries of biological imaging. These techniques enable visualization of cellular and molecular events at unprecedented resolution and depth, leading to a greater understanding of biological function and dysfunction. The rise of computational imaging methods, such as lens less microscopy and phase retrieval techniques, is enhancing the capabilities of biophotonic imaging. These methods often rely on advanced algorithms to reconstruct images from indirect measurements, overcoming limitations of traditional optics [200]. The integration of nano photonics and metasurfaces with biophotonics is opening up new avenues for imaging and sensing. Metasurfaces, in particular, offer the potential to create ultra-compact and highly functional optical components for biomedical applications [201]. Biophotonic probes, including fluorescent proteins, quantum dots, and other luminescent nanoparticles, are crucial tools for visualizing specific molecules and processes within living cells and tissues [202]. The field will continue to be advanced by both technological advancements and fundamental research. The integration of artificial intelligence and machine learning is poised to further revolutionize image analysis, diagnostics, and treatment planning. Developing more compact, portable, affordable, and user-friendly biophotonic devices will expand the accessibility and impact of these technologies in healthcare settings, especially in resource-limited areas. **A significant stride toward this goal is the work by** Niu et al. [203]**, who developed a hollow-microsphere-integrated optofluidic immunochip for the rapid microanalysis of a myocardial infarction biomarker, demonstrating a compact and sensitive platform ideal for point-of-care applications.** The analysis of the latest works evidences several key focus areas for progress.

6.1 Advanced microscopy techniques

A prominent area of research is advanced microscopy techniques, with numerous studies focused on developing and refining diverse methods such as fluorescence microscopy, light sheet microscopy, Raman microscopy, and optical coherence tomography. Notably, the integration of deep learning and various computational methods for enhancing image analysis and reconstruction is a consistent and growing trend in these advancements. Bruch et al. [204] enhance 3D deep learning segmentation of cells by employing biophysically motivated cell synthesis, emphasizing the use of simulated cell data to improve the accuracy of deep learning models for cell segmentation. [77, 111, 205–207] utilize deep learning to enhance light sheet fluorescence microscopy for in vivo 4D imaging of zebrafish heart beating, demonstrating the synergistic combination of advanced microscopy and deep learning for improved dynamic in vivo imaging in biological systems. [126] develop fluorescence intensity and lifetime imaging techniques for a viscosity-sensitive probe to study cell membrane tension and apoptosis processes at the cellular level. [67, 68] develop isotropic 3D microscopy using weakly physics-informed, domain-shift-resistant axial deblurring methods to enhance image quality in 3D microscopy by reducing blurring and artifacts. [208] review machine learning-empowered coherent Raman imaging and analysis for biomedical applications, highlighting the synergistic combination of coherent Raman microscopy and machine learning for improved biomedical imaging and data analysis. Oh et al. [209] describe digital aberration correction for enhanced thick tissue imaging by exploiting aberration matrix and tilt-tilt correlation from the optical memory effect, advancing microscopy image quality in thick tissues by utilizing optical memory effects and computational aberration correction. Additional contributions include: Surkov et al. [210] developing laser speckle contrast imaging with principal component and entropy analysis, introducing a novel approach for depth-independent blood flow assessment,Li Y., Zhang, C., et al. (2025) developing computational multi-angle optical coherence tomography using implicit neural representation; and Narotamo et al. [211] introducing 3DCellPol, a method for joint detection and pairing of cell structures to compute cell polarity.

6.2 In Vivo imaging

Another significant emphasis lies on in vivo imaging, with a considerable number of references demonstrating the application of biophotonic techniques for the direct study of biological processes within living organisms. Chang et al. [212] provide a comprehensive review of in vivo surface-enhanced Raman scattering (SERS) techniques, encompassing nanoprobes, instrumentation advancements, and diverse biomedical applications of this spectroscopic method. The enhancement factor (EF) in SERS is a measure of the amplification of the Raman signal and can be approximated as:

$$EF \approx |E1oc\left(\omega L\right)/Einc\left(\omega L\right)\ |^4$$

Where Eloc and Einc are the local and incident electric fields at the laser frequency ωL, respectively. [77, 111, 205, 206, 208] utilize deep learning to enhance light sheet fluorescence microscopy for in vivo 4D imaging of zebrafish heart beating. Lin et al. (2025) develop dye-sensitized nanocomposites for in vivo near-infrared (NIR)-IIb vascular and tumor imaging, demonstrating the use of novel nanomaterials for enhanced in vivo imaging in the NIR-IIb window for vascular and tumor visualization. Gangadaran et al. [213] review noninvasive in vivo imaging techniques for macrophages, focusing specifically on methods for imaging macrophages in living systems without invasive procedures, essential for studying immune responses in vivo. [214] report advancements in PMN-PT transparent piezoelectric ceramic for photoacoustic and ultrasound dual-mode imaging, highlighting material innovations for combined photoacoustic and ultrasound imaging modalities, enhancing multimodal biomedical imaging capabilities. Additional contributions include: Seo et al. [215] exploring the 3D scanner's potential as a novel tool for lymphedema measurement in mouse hindlimb models, showcasing a new application of 3D scanning technology for quantitative assessment of lymphedema in animal models.

6.3 Nanomaterials for bioimaging and therapy

The utilization of nanomaterials for bioimaging and therapy is a recurring theme, with several papers highlighting the diverse applications of nanomaterials like upconversion nanoparticles, quantum dots, and metal–organic frameworks in bioimaging, targeted drug delivery, and various therapeutic interventions. Kaplan and Uzun [216] review the biomedical applications of upconversion nanoparticles, with a particular focus on their utility in cancer diagnosis and monitoring telomere activity in biological systems. Lin et al. (2025) develop dye-sensitized nanocomposites for in vivo near-infrared (NIR)-IIb vascular and tumor imaging. Al-Mashriqi et al. [217] develop a nanoprobe for monitoring pepsin activity and cell imaging, creating a fluorescent nanoprobe designed for real-time monitoring of pepsin enzymatic activity and cellular imaging applications. Babayevska et al. [218] develop doxorubicin and ZnO-loaded gadolinium oxide hollow spheres for combined cancer therapy and bioimaging, creating a multifunctional nanoparticle platform for simultaneous drug delivery and bioimaging for cancer treatment applications. Chandra et al. [219] review smart nano-hybrid metal–organic frameworks for biomedical applications, providing a comprehensive overview of the applications and advancements of metal–organic frameworks in diverse biomedical contexts. Kohli and Parab [220] provide a comprehensive insight into the green synthesis of carbon quantum dots and their diverse applications, reviewing sustainable methods for producing carbon quantum dots and their wide-ranging applications in various fields. Adel et al. [221] review the biomedical applications of Boron Nitride Nanotubes, providing a broad overview of the diverse potential applications of Boron Nitride Nanotubes within the biomedical field. [155–157] describe the application of photoresponsive materials in bone regeneration, exploring the use of light-activated materials to promote and enhance bone tissue regeneration and healing processes. Additional contributions include [222] presenting high quantum yield AIE covalent organic frameworks for sensitive pH monitoring and copper toxicosis diagnosis and remission biosensors.

6.4 Medical image analysis and deep learning

Medical image analysis and deep learning form another critical theme, with the application of deep learning algorithms and other computational approaches for the analysis of medical images becoming increasingly prevalent. This includes a range of tasks such as image segmentation, feature extraction for diagnostic purposes, and automated disease diagnosis. **Complementing these computational advances in image analysis are innovations in sensor hardware, such as the dual-resonance optical fiber immunoprobe developed by** Dai et al. [135] **for the sensitive detection of the prostate cancer biomarker PSA, and the near-field enhanced plasmonic resonance immunoprobe for alpha-fetoprotein detection demonstrated by** Jing et al. [223]**.** Bruch et al. [204]

enhance 3D deep learning segmentation of cells by employing biophysically motivated cell synthesis. [113, 114] develop a targeted feature extraction model called TFEFusion for multimodal medical image fusion, highlighting the model's capability to effectively extract relevant features from different medical imaging modalities for improved image analysis. Rajesh et al. [224] employ machine learning to analyze retinal images and conclude that ethnicity is not strongly correlated with retinal pigment scores, suggesting that retinal pigment variations are not primarily determined by ethnicity according to their machine learning analysis. Aumente-Maestro et al. [225] present a deep learning approach, BTS U-Net, for brain tumor segmentation in medical images, showcasing the effectiveness of deep learning techniques, specifically BTS U-Net architecture, for accurate brain tumor segmentation. Alotaibi et al. [226] utilize deep learning for colorectal cancer diagnosis from biomedical images, showcasing the application of deep learning algorithms for automated colorectal cancer detection based on biomedical image analysis. Prakash et al. [227] utilize deep learning for cancerous tumor detection and classification from biomedical images, demonstrating the effectiveness of deep learning algorithms for automated detection and classification of cancerous tumors based on biomedical image analysis. Gu et al. [228] present DA-Net, a deep attention network designed for biomedical image segmentation, introducing a novel deep learning architecture specifically for improved segmentation accuracy in biomedical image analysis tasks. Additional contributions include: Ashame et al. [229] developing a computer-aided model for dental image diagnosis using convolutional neural networks,Chicco et al. [230] presenting a score for assessing the quality of biomedical datasets,and [83–86] developing a neural operator for solving partial differential equations.

6.5 Specific imaging modalities

Furthermore, several studies concentrate on specific imaging modalities, showcasing focused advancements in Positron Emission Tomography (PET), electrical impedance tomography, ultrasound imaging, and optical coherence tomography. [83–86] expand the range of tracers available for positron emission tomography (PET) by utilizing high molar activity Fluorine-18 (18F)-labeled compounds, thereby enhancing the capabilities and versatility of PET imaging in biomedical applications. McAteer et al. [231] provide recommendations for translating PET radiotracers for cancer imaging, focusing on the practical steps and guidelines for developing and translating new PET radiotracers into clinical applications for improved cancer imaging. Ren et al. [232] develop dynamic chest electrical impedance tomography with mixed statistical shape reconstruction, focusing on advancements in electrical impedance tomography techniques for dynamic chest imaging with improved reconstruction methods. Joshi et al. [233] review flexible micromachined ultrasound transducers (MUTs) for biomedical applications, focusing on the advancements and potential of MUT technology in both biomedical imaging and sensing applications. Menozzi et al. [234] present three-dimensional diffractive acoustic tomography as a novel imaging modality, highlighting the use of sound waves to achieve 3D imaging capabilities for potential biomedical applications. [214] report advancements in PMN-PT transparent piezoelectric ceramic for photoacoustic and ultrasound dual-mode imaging. [235] develop computational multi-angle optical coherence tomography using implicit neural representation, advancing OCT imaging through computational methods and neural networks to achieve multi-angle imaging. Additional contributions include: Kanno et al. [236] introducing a high-throughput fluorescence lifetime imaging (FLIM) flow cytometry system,Mattana et al. [237] evaluating safe exposure times for two-photon microscopy imaging of painted surfaces,and [118, 119] developing broad-band multi-distance lensless imaging.

Figure 8 presents the FLIM flow cytometry system and its functionality: (a) A schematic diagram of the cytometer setup. (b) The fluorescence lifetime acquisition method using modulation frequencies. (c) Amplitude and phase images generated from modulated beams. (d-g) FLIM flow cytometry images of different samples, including polymer beads, Euglena gracilis, rat glioma cells, and Jurkat cells. This system highlights the potential of FLIM-based flow cytometry in biomedical research.

6.6 Theragnostic and metalens applications

The integration of theragnostic and therapy is also represented, with research combining diagnostic and therapeutic functionalities into single platforms for more effective and targeted healthcare solutions. **A prime example of this is the work by** [131] **who**

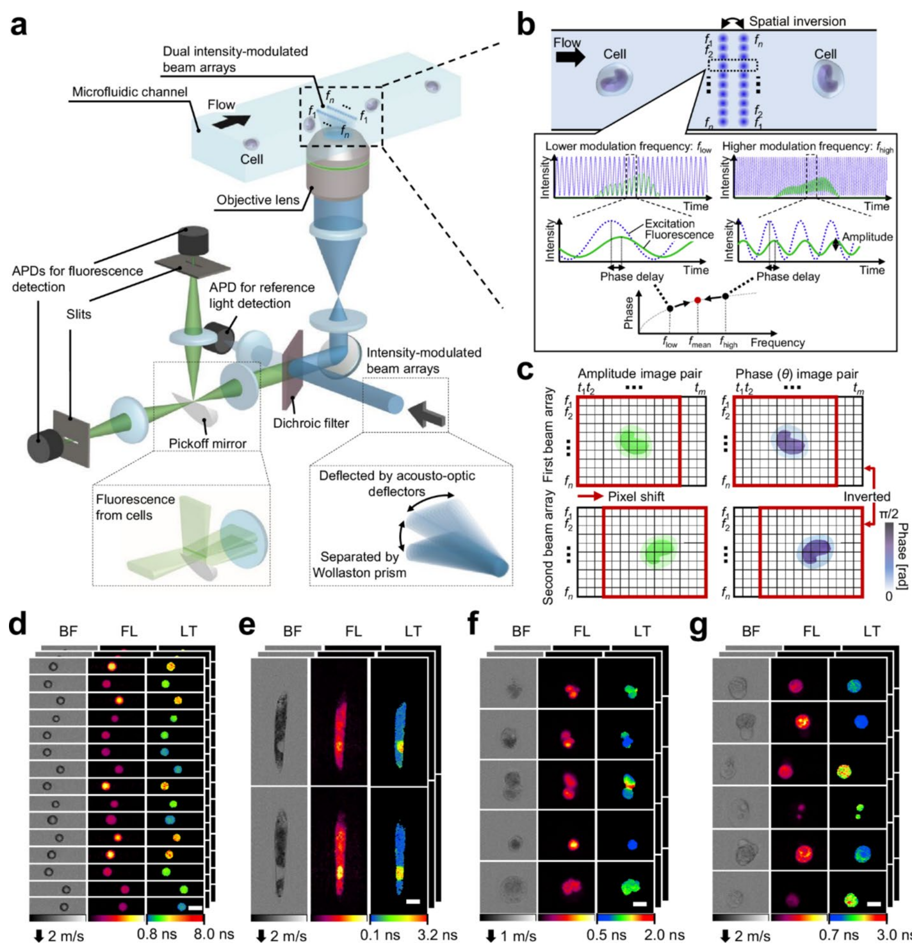

Fig. 8 High-Throughput FLIM Flow Cytometry System. (**a**) Schematic of the cytometer with dual intensity-modulated beam arrays and APDs. (**b**) Principle of fluorescence lifetime pixel acquisition using complementary modulation frequencies (f_{low}, f_{high}) and mean frequency (f_{mean}). (**c**) Amplitude and phase image pairs from modulated beams. (**d**-**g**) FLIM flow cytometry images (BF, FL, LT): (**d**) Polymer beads with different lifetimes, (**e**) Euglena gracilis cells, (**f**) Rat glioma cells, (**g**) Human Jurkat cells (all SYTO16 stained). Scale bars: 10 μm. Image reproduced from [236]. Article published under a Creative Commons Attribution 4.0 International License http://creativecommons.org/licenses/by/4.0/

developed multi-dimensional donor-engineered NIR-II AIEgens for multimodal photo theragnostics, demonstrating a sophisticated nanomaterial platform capable of both imaging and treating orthotopic breast cancer. This concept has been extended to the analysis of extracellular vesicles, key cancer biomarkers, through the use of autonomous microlasers developed by [238], **showcasing a sophisticated photonic probe for both detection and profiling from complex biological samples.** [218] develop doxorubicin and ZnO-loaded gadolinium oxide hollow spheres for combined cancer therapy and bioimaging. [222] present high quantum yield AIE covalent organic frameworks for sensitive pH monitoring and copper toxicosis diagnosis and remission biosensors. Finally, metalenses, as compact and increasingly versatile optical components, are also emerging in biomedical applications, indicating their growing relevance within the field of biophotonics and biomedical imaging. Yao et al. [6] develop a nonlocal Huygens' meta-lens for high-quality-factor spin-multiplexing imaging, demonstrating the application of metalenses to achieve advanced imaging modalities, including spin-multiplexing, with high image quality. [77, 111, 205–207] design a broadband terahertz achromatic metalens using deep learning, showcasing the use of deep learning to optimize the design of terahertz metalenses capable of achromatic focusing over a broad bandwidth, potentially relevant to future biomedical terahertz imaging applications. Chen J. et al. (2025) present a 3D-printed aberration-free terahertz metalens for ultra-broadband achromatic super-resolution wide-angle imaging with high numerical aperture, demonstrating the creation of a high-performance terahertz metalens using 3D printing for advanced imaging capabilities at terahertz frequencies. Additional contributions include: [239] presenting a minimally invasive animal model of atherosclerosis and neointimal hyperplasia,Rowe et al. [240] reviewing human perception of ionizing radiation [187, 188] performing a bibliometric analysis of research trends in pilocytic astrocytoma,Flores-Aguilar et al. [241] presenting proceedings of a conference focused on Trisomy 21 research, [242] reviewing research on pediatric mild traumatic brain injury,Deepthy et al. [243] analyzing a three-layered antenna specifically designed for biomedical applications,Karadağlı and Uğur Aydın [244] comparing irrigation activation systems in endodontics using lasers,Gosée et al. [245] reviewing contrast enhancement techniques in medical imaging [5, 112, 132, 180, 246, 247] linking altered neuronal and synaptic properties to nicotinic receptor Alpha5 subunit gene dysfunction,and [248] presenting programmed polymeric integration of gold nanoparticles into multi-material 3D printed hydrogels.

A prominent area of research is advanced microscopy techniques, with numerous studies focused on developing and refining diverse methods such as fluorescence microscopy, light sheet microscopy, Raman microscopy, and optical coherence tomography [67, 68, 77, 111, 126, 204–209]. Notably, the integration of deep learning and various computational methods for enhancing image analysis and reconstruction is a consistent and growing trend in these advancements. Another significant emphasis lies on in vivo imaging, with a considerable number of references demonstrating the application of biophotonic techniques for the direct study of biological processes within living organisms [77, 111, 144, 205–207, 212–214]. The utilization of nanomaterials for bioimaging and therapy is a recurring theme, with several papers highlighting the diverse applications of nanomaterials like upconversion nanoparticles, quantum dots, and metal–organic frameworks in bioimaging, targeted drug delivery, and various therapeutic interventions [126, 144, 155–157, 216–221].

Medical image analysis and deep learning form another critical theme, with the application of deep learning algorithms and other computational approaches for the analysis of medical images becoming increasingly prevalent [113, 114, 204, 211, 224–228]. This includes a range of tasks such as image segmentation, feature extraction for diagnostic purposes, and automated disease diagnosis. **Complementing these computational advances in image analysis are innovations in sensor hardware, such as the dual-resonance optical fiber immunoprobe developed by** Dai et al. [135] **for the sensitive detection of the prostate cancer biomarker PSA, and the near-field enhanced plasmonic resonance immunoprobe for alpha-fetoprotein detection demonstrated by** [223]**.** Furthermore, several studies concentrate on specific imaging modalities, showcasing focused advancements in Positron Emission Tomography (PET) [83–86, 231], electrical impedance tomography [214, 232–234], and optical coherence tomography [249]. The integration of theragnostic and therapy is also represented, with research combining diagnostic and therapeutic functionalities into single platforms for more effective and targeted healthcare solutions [218, 222]. **A prime example of this is the work by** [131] **who developed multi-dimensional donor-engineered NIR-II AIEgens for multimodal photo theragnostics, demonstrating a sophisticated nanomaterial platform capable of both imaging and treating orthotopic breast cancer. This concept has been extended to the analysis of extracellular vesicles, key cancer biomarkers, through the use of autonomous microlasers developed by** [238]**, showcasing a sophisticated photonic probe for both detection and profiling from complex biological samples.** Finally, metalenses, as compact and increasingly versatile optical components, are also emerging in biomedical applications, indicating their growing relevance within the field of biophotonics and biomedical imaging [6, 77, 111, 205–207],and [148]. In conclusion, this collection of recent publications clearly illustrates the vital role of biophotonics and biomedical imaging in driving advancements in healthcare. The research presented spans a broad array of techniques, from sophisticated microscopy and in vivo imaging to the development of innovative nanomaterials and computational methodologies for enhanced image analysis. The synergistic integration of these diverse technologies is propelling significant progress in disease diagnosis, treatment monitoring strategies, and fundamental biological research.

This collection of references strongly underscores the vibrant and impactful field of biophotonics and biomedical imaging, alongside the crucial role of related computational techniques in advancing this domain. Several key themes are clearly evident within these studies. Table 4 and Fig. 9 summarize the key focus areas in biophotonics and biomedical imaging.

7 Sustainability and green photonics

The growing global awareness of environmental challenges, including climate change, resource depletion, and pollution, has made sustainability a central theme across all scientific and technological disciplines [250]. Photonics, with its inherent energy efficiency and potential for resource-efficient solutions, is increasingly recognized as having a crucial role to play in creating a more sustainable future [251, 252]. "Green Photonics," a term that has gained traction in recent years, encompasses the development and

Table 4 Summary table of key focus areas in biophotonics and biomedical imaging

Key area	Description
Advanced microscopy techniques	Development and improvement of microscopy techniques like fluorescence, light sheet, Raman, and OCT, often integrating deep learning and computational methods for image enhancement and analysis
In Vivo imaging	Focus on biophotonic techniques for studying biological processes directly within living organisms, enabling real-time and dynamic observation of physiological and pathological events
Nanomaterials for bioimaging and therapy	Utilization of nanomaterials (up conversion nanoparticles, quantum dots, MOFs) for diverse biomedical applications including enhanced bioimaging, targeted drug delivery, and various therapeutic interventions
Medical image analysis and deep learning	Application of deep learning and computational methods for analyzing medical images for tasks like segmentation, feature extraction, disease diagnosis, and improved image interpretation in clinical settings
Specific imaging modalities	Focused research and advancements in specific imaging modalities including PET, electrical impedance tomography, ultrasound imaging, and optical coherence tomography, pushing the boundaries of each modality's capabilities
Theragnostic and therapy	Development of integrated platforms that combine diagnostic and therapeutic functionalities, enabling simultaneous disease diagnosis and targeted treatment for enhanced healthcare outcomes
Metalenses in biomedicine	Emerging application of compact metalenses as optical components in biomedical imaging devices, showcasing their potential for miniaturization and advanced functionalities in biophotonics

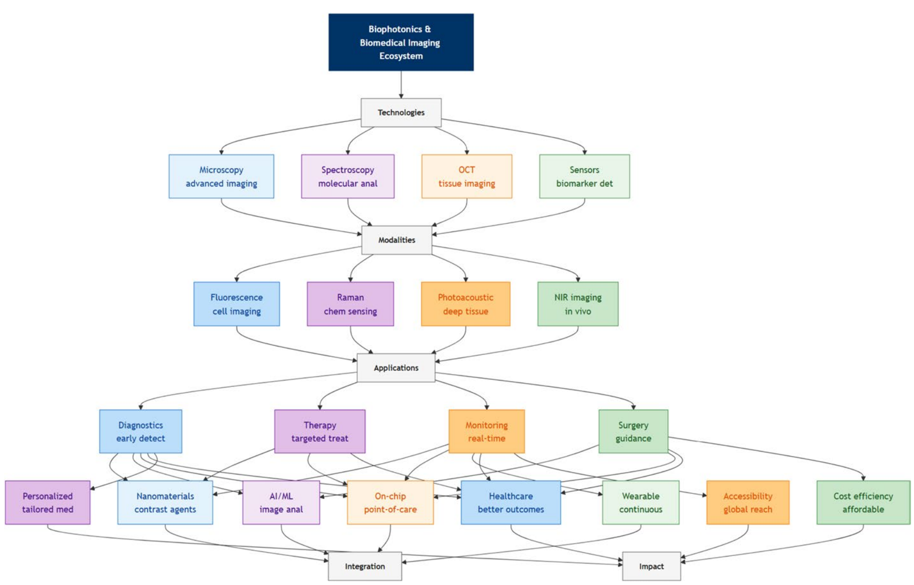

Fig. 9 Biophotonics and biomedical imaging ecosystem connecting imaging technologies, modalities, clinical applications, integration platforms, and healthcare outcomes

application of photonic technologies that minimize environmental impact and promote sustainability across various sectors [253]. This includes not only reducing the energy consumption of photonic devices themselves but also leveraging photonics to enable more sustainable practices in other areas. Photonics is inherently energy-efficient. Light-emitting diodes (LEDs) have already revolutionized lighting, offering substantial energy savings compared to traditional incandescent and fluorescent lighting. The efficiency of an LED is often characterized by its external quantum efficiency (EQE), which is the ratio of the number of photons emitted to the number of electrons injected. It is given by:

$$EQE = \eta_{inj} \times \eta_{rad} \times \eta_{ext}$$

Where η_{inj} is the injection efficiency, η_{rad} is the radiative efficiency, and η_{ext} is the light extraction efficiency.

Further advancements in solid-state lighting, as well as in optical communication networks that rely on low-loss optical fibers, are crucial for reducing global energy consumption ([251]). Photonic sensors are vital for monitoring air and water quality, detecting pollutants, and tracking climate change. Remote sensing, using technologies like LiDAR and hyperspectral imaging, provide crucial data for managing and protecting our environment [254]. The development of "green photonic biosensing" methods is also highlighted for its potential in point-of-care diagnostics, using sustainable materials and processes [254]. The materials used in photonic devices are important from an environmental and ethics perspective. Several approaches contribute. One is bio-inspired and includes utilizing natural materials like silk and paper, offering biodegradable and renewable alternatives to traditional electronic and photonic components [255]. In the realm of renewable energy, the power conversion efficiency (PCE) of a solar cell is a key metric for its performance. The ultimate theoretical efficiency of a single p–n junction solar cell is defined by the Shockley-Queisser Limit. This limit accounts for the trade-off between absorbing a maximum number of photons and minimizing thermalization losses. For a single-junction cell under standard solar illumination, the maximum theoretical efficiency is approximately $\eta_{max} \approx 33.7\%$. The PCE is defined as:

$$PCE = (J_{SC} \times V_{oc} \times FF)/P_{in}$$

Where J_{sc} is the short-circuit current density, V_{oc} is the open-circuit voltage, FF is the fill factor, and P_{in} is the incident optical power.

Sustainable fabrication remains an important challenge for green photonics, requiring holistic sustainability metrics in photonic manufacturing.

Using less material is also relevant to greener solutions. Vertical-cavity surface-emitting lasers (VCSELs) and other nanophotonic devices exemplify this approach, minimizing material usage while maximizing efficiency [256]. The manufacturing of photonic components, like any technology, must minimize waste, avoid harmful substances, and strive for a circular economy approach. Green photonics emphasizes sustainable design for long lifecycles and reduced need for resource inputs [253]. The future of green photonics hinges on several key areas: developing new, environmentally friendly materials for photonic devices,designing and manufacturing photonic systems with reduced energy consumption and material usage; and expanding the application of photonics in areas that promote sustainability, such as renewable energy, environmental monitoring, and precision agriculture. The field is driven by innovation in core photonics, as well as materials science and engineering to ensure alignment with sustainable development goals.

7.1 Integrated photonics and silicon photonics

A central theme that emerges is the relentless progress in integrated photonics and silicon photonics. Several studies illuminate this area, spanning the development of advanced optical switches, the creation of innovative integrated lasers and amplifiers, explorations into in-memory computing architectures, and the realization of reconfigurable optical logic functionalities. These endeavors collectively underscore the ongoing pursuit of miniaturization, enhanced integration density, and superior functionality within photonic circuits. [257] provide an invited review paper summarizing the research progress in silicon optical switching devices, crucial components for optical networks and data centers. [30] demonstrate the fabrication of titanium:sapphire-on-insulator integrated lasers and amplifiers, marking the first realization of integrated photonic devices based on this material platform with potential performance advantages. [258] present a review paper focusing on in-memory computing devices and integrated chips that leverage chalcogenide phase change materials for efficient AI and data-intensive computing. Xia et al. [259] present a seven-bit nonvolatile electrically programmable photonics device based on phase-change materials, demonstrating its suitability for optical in-memory computing for image recognition tasks. [260, 261] present optically-reconfigurable integrated optical directed logic computing based on silicon photonics, demonstrating a silicon photonic chip capable of reconfigurable logic operations using optical control for optical computing. Additional contributions include: He et al. [262] developing selective area grown photonic integrated chips that completely suppress the Stokes shift in Raman spectroscopy,Lanziano et al. [263] developing a 1×8 optical splitter based on polycarbonate multicore polymer optical fibers,and Shah et al. [264] demonstrating mode-division multiplexing for visible photonic integrated circuits.

7.2 New materials for photonics

Complementing the drive towards integration is a vigorous exploration of new materials for photonics applications. A significant body of work is dedicated to characterizing and harnessing the potential of diverse material systems, ranging from phase-change materials and two-dimensional materials such as MXenes and ReS2, to carbon dots, covalent organic frameworks, metal halide perovskites, and an array of rare-earth doped materials. The overarching objective is to discover and refine materials possessing superior optical properties, enhanced tunability, and novel functionalities tailored for next-generation LEDs, lasers, sensors, and a host of other advanced photonic devices. [265] explore the utilization of greenly synthesized titanium dioxide nanoparticles (TiO2 NPs) to modulate the dielectric and electro-optical properties of nematic liquid crystals (NLCs), suggesting their potential in display applications by effectively altering the material characteristics. [266] study two-dimensional MXenes flakes exhibiting large second harmonic generation (SHG) and unique surface responses, highlighting the strong nonlinear optical properties of MXenes for nonlinear photonics. [267] develop an active pixel image sensor array based on large-scale ReS2 semiconducting film, exploring 2D semiconductors for advanced image sensors in optoelectronics and imaging technologies. [268] characterize amphiphilic acetylacetone-based carbon dots, focusing on the synthesis and properties of carbon dots with amphiphilic nature for potential photonic applications. [269] study cerium-doped pyrimidine-based covalent organic frameworks (COFs) for their photoluminescence, photocatalysis, and electrical properties, exploring COFs as materials with tunable optical and electrical characteristics. [270] study the effect of water molecules position in an eco-friendly Mn-based organic–inorganic metal halide material for LEDs and solvent vapor sensing, exploring metal halide materials for LEDs and sensing with a focus on environmental factors. [271] investigate the wide-range emission properties of Er3 + /Yb3 + co-doped lanthanum niobates for biophotonics applications, examining their temperature and excitation-dependent luminescence for potential biomedical uses. [272] study the impact of synthesis methods and coordination agents on the structure, morphology, and luminescent efficiency of triple-doped yttrium orthovanadate phosphors, optimizing phosphor synthesis for enhanced performance. [273] study temperature-dependent photoluminescence of rare-earth doped LaCa4O(BO3)3, characterizing luminescent materials for lighting, displays, and optical thermometry. Additional contributions include [274] studying ultrafast shift current in SnS2 single crystals and its terahertz emission,and [275] studying leveraging femtosecond laser ablation for tunable near-infrared optical properties in MoS2-gold nanocomposites.

7.3 Luminescent materials and applications

A substantial portion of the presented research centers on luminescent materials and their multifaceted applications. Numerous investigations delve into the synthesis, detailed characterization, and practical deployment of luminescent materials, with a focus on their use in applications ranging from advanced LEDs and sophisticated bio-imaging techniques to high-performance sensors and cutting-edge optical data storage solutions. Within this domain, key performance indicators include achieving tunable luminescence characteristics and maximizing overall luminous efficiency. **A comprehensive overview of the state-of-the-art in this area is provided by Zhang T. et al.** [77]**, who review the status and perspective of inorganic ligand-capped quantum dot light-emitting diodes (QLEDs), a key technology for next-generation, energy-efficient displays and lighting.** Yu et al. [276] present a novel 0D Mn(II)-based hybrid halide nanofiber film exhibiting ultra-stable, multimodal, and reversible luminescence switching, suggesting its potential utility in various photonic applications. [65, 66] demonstrate robust low threshold full-color upconversion lasing in a rare-earth activated nanocrystal-in-glass microcavity, showcasing a compact and efficient method for generating full-color laser light. [58, 59] study fluorescence enhancement of a charge-transfer cocrystal and hydrolysis-activated cocrystal-to-cocrystal transformation, revealing how cocrystal transformation can be used to modulate fluorescence properties. Qi et al. [277] develop one-dimensional molecular co-crystal alloys capable of full-color emission for low-loss optical waveguides and optical logic gates, presenting new color-tunable materials for integrated photonics applications. Abeywickrama et al. [278] trap-engineer the persistent luminescence of Ca3Ga4O9:Tb3 + via Al3 + substitution for optical data storage, enhancing material properties for optical data storage applications based on persistent luminescence. [260, 261] molecularly engineer efficient aggregation-induced delayed fluorescence (AIDF) luminogens for high-performance OLEDs, focusing on materials for OLEDs to improve display and lighting technologies. Rakov et al. [279] develop Mn2 + -doped Zn2SiO4 phosphors for optical thermometry in the visible region, showcasing a new phosphor material for temperature measurement using optical techniques and directly contributing to optical thermometry. Additional contributions include: Senica et al. [280] publishing a minor correction to a previously released paper concerning terahertz quantum cascade lasers,Torun et al. [281] investigating coffee-ring mediated thinning and thickness-dependent dewetting modes in printed polymer droplets, coupled with the assembly of quantum dots, for anti-counterfeiting applications,and

[39, 73, 97, 158, 168, 178, 282] designing and applying chirped mirrors in the visible light band.

7.4 Metamaterials and plasmonics

Metamaterials and plasmonics remain highly active and productive research frontiers. Ongoing work includes the development of novel metasurface fabrication methodologies, the creation of tunable retroreflectors with surface plasmon mediation, the realization of flexible plasmonic surface-enhanced Raman spectroscopy (SERS) sensors, and the application of metamaterials in advanced image processing systems. Looking ahead, the potential role of metamaterials in catalysis is also being considered, broadening their scope beyond traditional optical applications. Blair et al. [283] present a "green" etching method for fabricating indium tin oxide (ITO) metasurfaces, emphasizing an environmentally conscious approach to metasurface nanofabrication. [284] demonstrate surface plasmon mediated angular and wavelength tunable retroreflectors using parallel-superimposed surface relief bi-gratings, showcasing tunable optical components based on surface plasmon effects. [285] develop a flexible 3D plasmonic web that enables remote surface-enhanced Raman spectroscopy (SERS), showcasing a flexible plasmonic sensor for remote detection applications in plasmonic and biophotonics. [286] develop a reconfigurable nonlocal thin film nano-cavity for image processing, showcasing a tunable nano-cavity structure for advanced optical image processing applications. Loh [287] provides a perspective article questioning the potential of metamaterials becoming important in catalysis, exploring broader applications of metamaterials beyond traditional optics. Additional contributions include: Choudhury and Gric [288] providing an overview of metamaterials, covering their design principles and applications, and discussing both ordered and disordered types of metamaterials,and Rezzouk et al. [289] studying Fabry-Pérot and Friedrich-Wintgen bound states in the continuum (BIC) in a photonic triple-stub cavity.

7.5 Biophotonics and sensing

The domain of biophotonics and optical sensing is significantly represented, with research articles dedicated to optical biosensors designed for ketone body detection, flexible photonic pressure sensors, optical thermometry based on novel materials, and remote SERS sensing enabled by flexible plasmonic structures. Further enriching this area, luminescent materials tailored for biophotonic applications are also being actively explored and refined. [290] develop β-Hydroxybutyrate dehydrogenase functionalized two-dimensional photonic crystals for quantitative and visual sensing of ketone bodies, presenting a photonic crystal biosensor for diabetes monitoring. Fang et al. [291] develop a pressure visualization and quantification photonic skin based on a flexible optical fiber combiner, showcasing a flexible pressure sensor based on optical fiber technology. [279] develop Mn^{2+}-doped Zn_2SiO_4 phosphors for optical thermometry in the visible region. Rodríguez-Sevilla et al. [285] develop a flexible 3D plasmonic web that enables remote SERS. Manfré et al. [271] investigate the wide-range emission properties of Er^{3+}/Yb^{3+} co-doped lanthanum niobates for biophotonics applications. [136] develop ethanol-responsive structural colors with multi-level information encryption based on patterned inverse opal photonic crystals, exploring photonic crystals for sensing and security applications through responsive materials. Additional contributions include: Esposito et al. [292] providing an editorial overview of a special issue focused on recent advancements in optical biosensors and chemical sensors,and [293] developing a simple sol–gel protein stabilization method for rainbow and white lighting devices.

7.6 Sustainable photonics and green fabrication

Reflecting a growing global awareness, sustainable photonics and green fabrication are emerging as critical considerations within the field. This is evidenced by investigations into environmentally conscious etching techniques, broad perspectives on sustainable photonics encompassing diverse applications, and the development of environmentally responsible materials processing methodologies. [283] present a "green" etching method for fabricating indium tin oxide (ITO) metasurfaces. [294] discuss sustainable photonics for energy harvesting and beyond in a perspective article, covering solar cells, OLEDs, communication, and smart windows and addressing sustainability in photonics applications. [295] present an environmentally conscious and highly effective method for sorting single-walled carbon nanotubes (SWCNTs) using recurrent conjugated polymer extraction, addressing sustainable materials processing for carbon nanotubes. [7] present their work on developing bio-sourced flexible supramolecular glasses exhibiting dynamic and full-color phosphorescence. They created these glasses using bio-based materials, likely proteins and other organic compounds, to form flexible glass matrices capable of long-lived phosphorescence. The emission color and properties of these glasses can be tuned. The paper details the preparation methods of these bio-glasses, the characterization of their structural and phosphorescent properties, and an investigation into the mechanism behind their long-lived and tunable phosphorescence. This research opens up new avenues for sustainable and functional optical materials based on bioresources, with potential applications in areas like displays, sensors, and security. Additional contributions include: [296] reviewing microbial nanotechnology for precision nano biosynthesis, highlighting innovations, opportunities, and future perspectives for industrial sustainability in nanomaterial production,and [297] developing flexible composites from water-dispersible components, including polyvinyl alcohol, Janus nanoparticles, and polyaniline, towards mixed ionic-electronic conductors with relevance to flexible photonics and optoelectronics.

Figure 10 presents the preparation, room temperature phosphorescence (RTP), and mechanism analysis of their bio-based glasses. Panel (a) illustrates the preparation process of E–G and PCA-doped glasses, showing samples at room temperature and proposing a mechanism for the long-lived phosphorescence observed in these glasses. Panel (b) displays photographs of the glasses after exposure to UV light, demonstrating their phosphorescent emission. Panel (c) shows docking simulation results from AutoDockTools, visualizing the interaction of ovalbumin (OVA) within an EA glass matrix and a partial peptide chain (PPC) within a GEL glass matrix. Scale bars are included in panels (a) and (b).

7.7 Nonlinear optics and frequency conversion

Finally, the fundamental aspects of nonlinear optics and frequency conversion are addressed through studies on second harmonic generation in lithium tantalate thin films and the characterization of second harmonic generation in MXenes materials. [298] demonstrate continuous-wave second-harmonic generation of green light in periodically

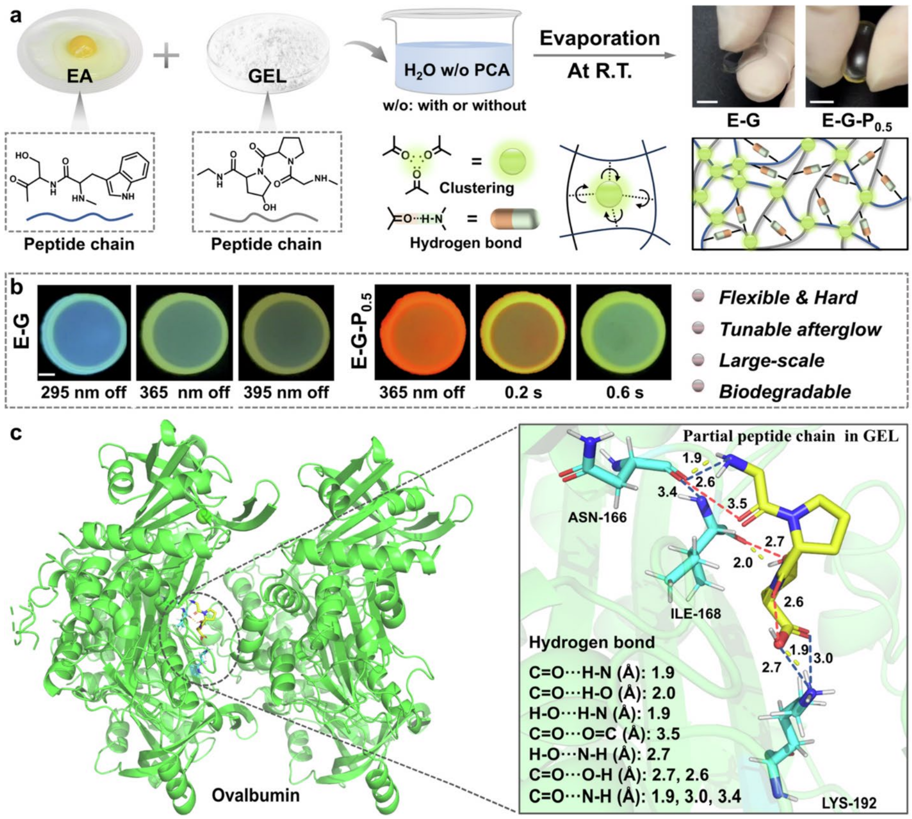

Fig. 10 Bio-based Glass Preparation, Phosphorescence, and Mechanism. (**a**) Preparation of E–G and PCA-doped glasses (scale bar: 3 mm) and proposed long-lived phosphorescence mechanism. (**b**) Photographs of glasses after UV exposure demonstrating phosphorescence (scale bar: 3 mm). (**c**) Docking simulation results (AutoDockTools): ovalbumin (OVA) in EA glass and partial peptide chain (PPC) in GEL glass. Image reproduced from [7]. Article published under a Creative Commons Attribution 4.0 International License http://creativecommons.org/licenses/by/4.0/

poled thin-film lithium tantalate (PPTLT), highlighting efficient frequency conversion for green light generation in integrated photonics. [266] study two-dimensional MXenes flakes exhibiting large second harmonic generation (SHG) and unique surface responses, highlighting the strong nonlinear optical properties of MXenes for nonlinear photonics. Additional contributions include [299] unraveling the nature of lasing emission from hybrid silicon nitride and colloidal nanocrystal photonic crystals with low refractive index contrast,and [300] elucidating the nanostructural organization and photonic properties of butterfly wing scales using hyperspectral microscopy, studying natural photonic structures for bio-inspired photonics.

A central theme that emerges is the relentless progress in integrated photonics and silicon photonics. Several studies illuminate this area, spanning the development of advanced optical switches [257], the creation of innovative integrated lasers and amplifiers [30], explorations into in-memory computing architectures [258, 259], and the realization of reconfigurable optical logic functionalities [260, 261]. These endeavors collectively underscore the ongoing pursuit of miniaturization, enhanced integration density, and superior functionality within photonic circuits. Complementing this drive towards integration is a vigorous exploration of new materials for photonics applications. A significant body of work is dedicated to characterizing and harnessing the potential of diverse material systems, ranging from phase-change materials [258, 259] and two-dimensional materials such as MXenes [266] and ReS2 [267], to carbon dots [268], covalent organic frameworks [269], metal halide perovskites [270], and an array of rare-earth doped materials [271–273]. The overarching objective is to discover and refine materials possessing superior optical properties, enhanced tunability, and novel functionalities tailored for next-generation LEDs, lasers, sensors, and a host of other advanced photonic devices.

A substantial portion of the presented research centers on luminescent materials and their multifaceted applications. Numerous investigations delve into the synthesis, detailed characterization, and practical deployment of luminescent materials [58, 59, 65, 66, 260, 261, 268, 270, 272, 273, 276–279], with a focus on their use in applications ranging from advanced LEDs and sophisticated bioimaging techniques to high-performance sensors and cutting-edge optical data storage solutions. Within this domain, key performance indicators include achieving tunable luminescence characteristics and maximizing overall luminous efficiency. **A comprehensive overview of the state-of-the-art in this area is provided by** [77]**, who review the status and perspective of inorganic ligand-capped quantum dot light-emitting diodes (QLEDs), a key technology for next-generation, energy-efficient displays and lighting.** Metamaterials and plasmonics remain highly active and productive research frontiers. Ongoing work includes the development of novel metasurface fabrication methodologies [283], the creation of tunable retroreflectors with surface plasmon mediation [284], the realization of flexible plasmonic surface-enhanced Raman spectroscopy (SERS) sensors [285], and the application of metamaterials in advanced image processing systems [286]. Looking ahead, the potential role of metamaterials in catalysis is also being considered, broadening their scope beyond traditional optical applications [287].

The domain of biophotonics and optical sensing is significantly represented, with research articles dedicated to optical biosensors designed for ketone body detection (Zhou Y., Muhammad I., et al., 2024), flexible photonic pressure sensors [291], optical thermometry based on novel materials [279], and remote SERS sensing enabled by flexible plasmonic structures [285]. Further enriching this area, luminescent materials tailored for biophotonic applications are also being actively explored and refined [271]. Reflecting a growing global awareness, sustainable photonics and green fabrication are emerging as critical considerations within the field. This is evidenced by investigations into environmentally conscious etching techniques [283], broad perspectives on sustainable photonics encompassing diverse applications [294], and the development of environmentally responsible materials processing methodologies [295]. Finally, the fundamental aspects of nonlinear optics and frequency conversion are addressed through studies on second harmonic generation in lithium tantalate thin films [53] and the characterization of second harmonic generation in MXenes materials [266]. In summary, this collection of recent studies highlights a vibrant and multifaceted research environment within photonics. Progress in this field is consistently driven by innovations in materials science, advancements in nanofabrication capabilities, and the increasing power of computational design tools. These combined factors are enabling continuous breakthroughs in integrated photonics, optical computing architectures, advanced sensing technologies, efficient lighting solutions, and sustainable technological practices. The growing convergence of photonics with other scientific and engineering disciplines, such as biology and materials science, is also a defining characteristic of this dynamic field.

Table 5 Summary table of key focus areas in green photonics

Key area	Description
Integrated photonics and silicon photonics	Focus on miniaturization, integration, and enhanced functionality of photonic circuits, including optical switches, integrated lasers and amplifiers, in-memory computing, and reconfigurable optical logic, particularly in silicon photonics
New materials for photonics	Exploration and characterization of novel materials such as phase-change materials, 2D materials, carbon dots, COFs, metal halide perovskites, and rare-earth doped materials, aiming for improved optical properties, tunability, and functionalities
Luminescent materials and applications	Investigation of luminescent materials' synthesis, characterization, and applications in LEDs, bioimaging, sensors, and optical data storage, with a focus on tunable luminescence and enhanced efficiency
Metamaterials and plasmonic	Continued research in metamaterials and plasmonic, covering metasurface fabrication, tunable retroreflectors, flexible plasmonic SERS sensors, metamaterials for image processing, and exploring potential applications in catalysis
Biophotonics and sensing	Development of biophotonic sensors, pressure sensors, optical thermometry, and SERS-based remote sensors; exploration of luminescent materials for biophotonics applications, emphasizing applications in biomedicine and environmental monitoring
Sustainable photonics and green fabrication	Focus on environmentally conscious approaches in photonics, including green etching methods, sustainable photonic technologies in general (energy harvesting, efficient lighting), and environmentally responsible materials processing for photonics
Nonlinear optics and frequency conversion	Research into fundamental aspects of nonlinear optical phenomena and their applications, specifically studies on second harmonic generation in lithium tantalate thin films and MXenes for frequency conversion and advanced optical functionalities

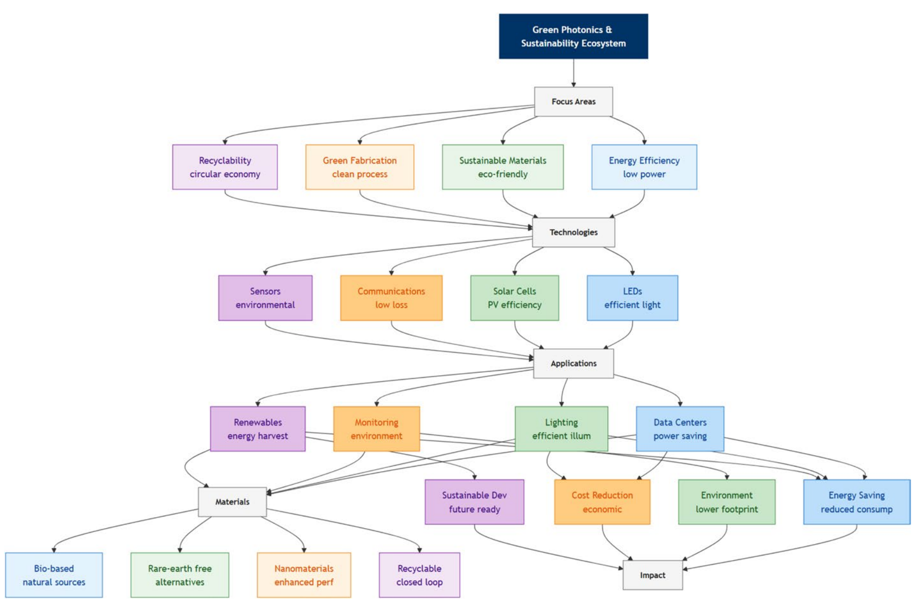

Fig. 11 Green photonics and sustainability ecosystem showing the focus areas, enabling technologies, applications, material innovations, and environmental and economic impacts

This collection of references encompasses a remarkable breadth of investigations across photonics and allied disciplines, effectively illustrating the dynamism and diversity of contemporary research. Table 5 and Fig. 11 summarize the key focus areas in green photonics.

8 Discussion and conclusion

The hot topics meticulously examined in this perspective, integrated photonics, quantum photonics, metaphotonics and metamaterials, biophotonics and biomedical imaging, and green photonics, collectively offer a vibrant and deeply insightful panorama of the dynamic and rapidly evolving landscape of photonics. Crucially, it is essential to recognize that these seemingly distinct domains are not operating in isolation; rather, they are increasingly interconnected and synergistic, with advancements in one area frequently acting as a catalyst for progress and innovation in others [301]. This interconnectedness is a hallmark of the current photonic revolution, fostering a rich ecosystem of cross-pollination and mutual reinforcement.

For instance, integrated photonics serves as a foundational platform, enabling the practical realization of compact, scalable, and robust quantum photonic systems. **A powerful demonstration of this synergy is the work by** Knaut et al. [302]**, who achieved the entanglement of nanophotonic quantum memory nodes directly within a telecom network, seamlessly blending integrated photonics, quantum memory, and fiber-optic infrastructure.** This integration is paramount for transitioning quantum technologies from specialized laboratory settings to widespread real-world applications through miniaturization and cost-effective mass production. Concurrently, the groundbreaking development of metamaterials offers entirely new paradigms for controlling and manipulating light at subwavelength scales, profoundly impacting both classical and emerging quantum applications. Metamaterials provide unprecedented optical functionalities, allowing for the design of devices with tailored responses and functionalities previously considered unattainable with conventional optical materials. Biophotonics, in turn, stands to gain significantly from advancements across these fields. Integrated photonics empowers the creation of increasingly miniaturized and highly sensitive biomedical sensors for point-of-care diagnostics and continuous health monitoring. Furthermore, the emergence of novel metamaterials enhances biomedical imaging by providing improved contrast mechanisms and enabling new functionalities for both diagnostic and therapeutic applications, such as targeted drug delivery and phototherapy.

Underpinning all these domains is the increasingly critical imperative of sustainability. Green photonics is not merely a sub-discipline but an overarching design principle that permeates the entire field. This sustainability focus profoundly influences material selection, driving the innovation of environmentally conscious fabrication processes, and shaping the fundamental design principles of photonic devices to minimize their environmental impact and maximize resource efficiency throughout their lifecycle.

Driven by a potent combination of fundamental scientific curiosity to unravel the intricate nature of light and its interactions with matter, coupled with the urgent demands of pressing societal needs for enhanced communication bandwidth, advanced healthcare solutions, and a sustainable environment, photonics is undeniably poised for continued exponential growth and transformative impact across diverse sectors. The ongoing convergence of photonics with other key disciplines, most notably artificial intelligence and machine learning, biotechnology, and advanced materials science, is generating exceptionally exciting and previously unforeseen opportunities.

To operationalize this convergence, we propose a forward-looking roadmap for cross-domain photonics innovation, aligned with unmet societal needs, focusing on the integration of metaphotonics with quantum sensors, scaling green photonics materials, and merging integrated photonics with biophotonics for advanced implantable monitoring devices.

Machine learning algorithms are revolutionizing the design and optimization processes for increasingly complex photonic structures and devices, enabling the realization of functionalities that were once considered beyond the reach of conventional design methodologies. The synergistic integration of photonics with biology is fundamentally reshaping healthcare, promising more precise and personalized diagnostics, minimally invasive therapies, and a deeper understanding of fundamental biological processes at the cellular and molecular level. Moreover, the continuous development of novel materials, including the fascinating realm of two-dimensional materials with their unique optoelectronic properties, quantum dots exhibiting tunable emission characteristics, and versatile organic compounds offering cost-effective and flexible device fabrication, is constantly expanding the horizons and capabilities of photonic devices and systems, promising significant performance enhancements and entirely new functionalities.

However, to fully realize the vast transformative potential of photonics and translate these exciting innovations into tangible real-world impact, it is crucial to address several key challenges that currently stand at the forefront of the field.

Key challenges and major bottlenecks include:

- **Achieving robust scalability and cost-effective manufacturability** of increasingly complex photonic devices, particularly when dealing with intricate three-dimensional structures or hybrid material integration. **For bandwidth scaling in communication systems, a fundamental bottleneck is the nonlinear Shannon limit of optical fibers, which restricts the maximum achievable data rate. Overcoming this requires advanced modulation formats, space-division multiplexing, or integrated photonic equalization circuits.**
- **Reducing energy consumption in photonic devices** remains paramount. **A critical bottleneck here is the thermal management in densely packed 3D integrated photonic circuits, where heat dissipation limits both performance and energy efficiency. The need for active cooling in high-density transceivers and switches directly counteracts the goal of green photonics.**
- **The relentless pursuit of novel materials** that possess specifically tailored optical properties, exhibit minimal optical losses across relevant spectral ranges, and maintain compatibility with existing fabrication techniques remains a high priority for continued advancement. **Specific material-related bottlenecks include the trade-off between photon purity and CMOS compatibility in quantum emitters, and the thermal conductivity limitations of heterogeneous integration platforms like silicon-lithium niobate.**
- **Minimizing the inherent energy consumption of photonic devices** is not only crucial for reducing operational costs, especially in large-scale deployments such as data centers and communication networks, but is also paramount for promoting overall sustainability and reducing the carbon footprint of technology.
- **Seamless integration of photonic components with electronic circuits** and other system elements, alongside the development of robust and reliable packaging solutions that can effectively protect delicate photonic devices and ensure stable performance under diverse real-world environmental conditions, is also essential for enabling practical and widespread applications beyond laboratory settings. **A key bottleneck limiting the widespread adoption of advanced photonics in telecommunications is the lack of standardized packaging and test protocols, which increases manufacturing costs and hinders scalability.**

- **Finally, cultivating and nurturing a highly skilled workforce**, one that is not only technically proficient in photonics but also adept at navigating the inherently interdisciplinary nature of the field and fostering collaborative innovation, is absolutely vital for sustaining continued progress and growth within this strategically important domain.

By actively fostering interdisciplinary collaborations that bridge across traditional scientific and engineering boundaries, by consistently investing in both fundamental research to expand our basic understanding of light-matter interactions and in the infrastructure necessary to translate fundamental discoveries into tangible technologies, and by diligently addressing these existing challenges in a concerted and collaborative manner, we can collectively and comprehensively harness the transformative power of light. Photonics is not simply illuminating the future in a metaphorical sense; it is actively and concretely shaping it, enabling a more interconnected, a more sustainable, and a healthier world for all.

Author contribution
Author: Writing – review and editing, Writing – original draft, Data curation, Conceptualization.

Funding
No funding was received for this work.

Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Generative AI and AI-assisted technologies
Generative AI and AI-assisted technologies have been used to improve the language and readability of the paper.

Competing interests
The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Received: 29 January 2026 / Accepted: 24 April 2026
Published online: 18 May 2026

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